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Inventor(s)	Sutter; Kevin J. et al.

Fluid management systems and methods

Abstract

Fluid management systems are disclosed that include software-controlled, electro-mechanical devices used in combination with single-use or multi-use tubing sets. Functions of the fluid management systems can include fluid pressurization, fluid warming, fluid deficit monitoring (including flow-based and weight-based), suction, fluid collection, and fluid evacuation (including indirect-to-drain and direct-to-drain options). The systems can be configured based on the surgical environment (e.g., operating room or physician office) as well as other user needs and/or preferences.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 62/932,921, filed Nov. 8, 2019.

TECHNICAL FIELD

(1) The present application is directed to fluid management systems and, more particularly, to fluid management systems and methods for surgical procedures.

BACKGROUND

(2) Surgical fluid management systems are used in endoscopic procedures to pressurize and deliver fluid to a surgical site for distending and continually flushing the surgical site to keep it free of blood and debris for visualization purposes.

(3) Fluid management systems may pressurize the fluid by manipulating the height from which fluid supply bags are hung relative to the height of the surgical site, controlling the air pressure in pressure cuffs or pressure chambers surrounding the fluid supply bags, or by pumping the fluid, typically with a peristaltic pump. Gravity provides non-pulsatile fluid flow, but poor fluid pressure control. Similarly, pressure cuffs or chambers provide non-pulsatile fluid flow, but poor fluid pressure control, unless the pressure in the cuffs or chambers is constantly adjusted to account for the fluid volume exiting the fluid supply bags. Peristaltic pumps can provide good pressure control, but the pulsatile nature of the fluid flow may impair distention and visualization at the surgical site.

(4) Fluid management systems may warm the fluid in order to assist in the mitigation or prevention of intraoperative hypothermia which can result in adverse outcomes. Such systems, however, may lack precise fluid temperature control, the ability to adequately warm fluid at the high fluid flow rates required for many procedures, and/or other capabilities required for certain surgical procedures (e.g., fluid deficit monitoring which is required for operative hysteroscopy). In facilities that lack fluid management systems with fluid warming capabilities, fluid bags may be pre-warmed in warming cabinets prior to use during a surgical procedure. However, use of such warming cabinets can result in dangerously hot fluid or, if the pre-warmed fluid is not used shortly after the pre-warming process has been completed, fluid that has cooled to room temperature and may contribute to intraoperative hypothermia.

(5) Fluid management systems may include a deficit monitoring system for calculating a deficit between an amount of a fluid supplied to a surgical site and an amount of fluid returned from a surgical site. Currently, fluid deficit monitoring is accomplished by subtracting the volume of fluid supplied to the surgical site from the volume of fluid returned from the surgical site into fluid collection canisters, bags, or vessels. The volume of fluid supplied is determined by monitoring the weight of the fluid source bags, counting the rotations of a peristaltic pump, and/or manually recording the number and volume of fluid bags utilized during the surgical procedure. The volume of fluid returned is determined by monitoring the weight of the canisters, bags, or vessels and/or manually viewing and recording the fluid levels in such canisters, bags, or vessels using graduation marks. In order for the returned fluid from the surgical site to move into the fluid collection canisters, the canisters are interconnected with tandem tubing and then connected to the surgical site and a suction source. If the fluid collection canisters become full during the surgical procedure, the procedure must be interrupted so that such canisters can be replaced. This process typically

involves suspending suction, disconnecting the canisters from the surgical site and suction source, disconnecting the tandem tubing, replacing the full canisters with new canisters, interconnecting the new canisters with tandem tubing, reconnecting the new canisters to the surgical site and suction source, and resuming suction. Due to the blood, tissue, and contaminated bodily fluids collected, full fluid collection canisters require regulated “red bag” disposal unless treated with solidifiers that, in most states, allow non-regulated “white bag” disposal.

(6) Fluid management systems may be connected to an internal or external suction source to pull fluid from a surgical site. As external suction sources are often set to high suction levels in an operating room environment, down-regulation may be necessary for proper operation of certain fluid outflow regulation, deficit monitoring, and/or collection functions. Down-regulation of an external suction source to provide desired suction levels may be accomplished via a manually or electronically controlled regulator. In order to isolate the regulator from biohazardous fluid, fluid collection canisters, bags, or vessels are typically placed between the regulator and the surgical site. These canisters, bags, or vessels must be removed and replaced after they become full during a surgical procedure.

(7) During endoscopic surgical procedures, a temporary increase in the fluid pressure and/or flow rate may be necessary to maintain or increase distention and/or to maintain or increase fluid flow for procedural and/or visualization purposes. To provide such a temporary increase, the user may manually operate a syringe, bulb, or similar device that is connected to the fluid inflow line of a surgical scope or instrument. As the duration of any increase in the fluid pressure and/or flow rate provided by these manual methods is limited to the volume of fluid contained in the syringe, bulb, or similar device, the necessary increase in the fluid pressure and/or flow rate may be interrupted while the syringe, bulb, or similar device is refilled with fluid. In other instances, the user may increase the fluid pressure by raising the height of the fluid supply bag, manually squeezing the fluid supply bag, or manually pumping up the pressure in a pressure bag or cuff surrounding the fluid supply bag. Alternatively, in some instances, the user may increase the setpoint fluid pressure of a fluid management system via a user interface such that the setpoint fluid pressure is set at a higher setpoint fluid pressure, and then the user decreases the setpoint fluid pressure to the original setpoint fluid pressure or other desired setpoint fluid pressure when the increase in pressure or flow is no longer necessary.

SUMMARY

(8) An exemplary embodiment of a fluid conditioner for a fluid management system includes a cartridge defining a first fluid chamber and a second fluid chamber. The first fluid chamber receives fluid from a fluid supply source and the second fluid chamber receives fluid from the first fluid chamber. At least a portion of the cartridge allows one or more non-contact sensors of the fluid management system to detect at least one characteristic of the fluid moving through the cartridge.

(9) An exemplary embodiment of a tubing set for a fluid management system includes a fluid conditioner, a first tube, and a second tube. The fluid conditioner includes a cartridge defining a first fluid chamber and a second fluid chamber, where the first fluid chamber receives fluid from a fluid supply source and the second fluid chamber receives fluid from the first fluid chamber. The second fluid chamber has an outlet for fluidly connecting the second fluid chamber to a surgical instrument. The first tube fluidly connects the fluid conditioner to the fluid supply source, and the second tube fluidly connects the fluid conditioner to the surgical instrument. At least a portion of the cartridge allows one or more non-contact sensors of the fluid management system to detect at least one characteristic of the fluid through the cartridge.

(10) An exemplary embodiment of a fluid management system includes a pump, a control system, and a fluid conditioning portion. The pump delivers fluid from at least one fluid supply container to a surgical site. The control system operates the pump and includes a user interface and one or more non-contact sensors. The user interface allows for communication between the control system and

the user, including inputting data to the control system. The one or more non-contact sensors detect one or more characteristics of the fluid being delivered to the surgical site. The fluid conditioning portion receives a fluid conditioner, and the fluid conditioner connects to the pump such that the fluid delivered from the pump moves through the fluid conditioner prior to moving to the surgical site.

(11) An exemplary method of monitoring fluid flow through a fluid management system includes configuring a control system of the fluid management system to detect a presence of the fluid at one or more location between the pump and the surgical site with one or more fluid presence sensors. The method further includes configuring the control system to provide a notification to a user through a user interface when at least one fluid presence sensor is not detecting fluid.

(12) An exemplary embodiment of a cartridge assembly for a fluid management system includes a fluid conditioner and a fluid warming cartridge. The fluid conditioner has a cartridge that defines a first fluid chamber and a second fluid chamber, where at least a portion of the cartridge allows one or more non-contact sensors of the fluid management system to detect at least one characteristic of the fluid moving through the cartridge. The fluid warming cartridge has a conduit that is fluidly connected to the first fluid chamber and the second fluid chamber of the fluid conditioner such that the first and second fluid chambers are fluidly connected. The fluid warming cartridge allows a heating source of the fluid management system to warm the fluid moving through the conduit when the fluid warming cartridge is used with the fluid management system.

(13) An exemplary embodiment of a cartridge assembly for a fluid management system includes a fluid conditioner and a pulse damping cartridge. The fluid conditioner has a cartridge that defines a first fluid chamber and a second fluid chamber, where the first fluid chamber receives fluid from a fluid supply source and the second fluid chamber receives fluid from the first fluid chamber. At least a portion of the cartridge allows one or more non-contact sensors of the fluid management system to detect at least one characteristic of the fluid moving through the cartridge. The pulse damping component includes at least one flexible sheet that at least partially defines a conduit, where the conduit fluidly connects the first fluid chamber and the second fluid chamber such that fluid received in the first fluid chamber from the fluid supply source moves through the conduit and into the second fluid chamber. The flexible side sheet expands and contracts as a pressure of the fluid moving through the conduit fluctuates to reduce pulsations of the fluid.

(14) An exemplary embodiment of a fluid management system includes at least one fluid supply container, a pump, and a control system. The pump delivers fluid from the fluid supply container to a surgical site. The control system operates the pump and includes a user interface for communication between the control system and a user, a bolus device operatively connected to the pump, and at least one processor operatively connected to the user interface, the bolus device, and the pump. The user interface allows a user to input at least one of a setpoint flow rate and a setpoint pressure of the fluid being delivered from the pump. The bolus device allows a user to increase at least one of the setpoint flow rate of the fluid to a temporary flow rate and the setpoint pressure of the fluid to a temporary pressure when the user activates the bolus device. The processor causes the pump to return to the setpoint flow rate and the setpoint pressure when the bolus device is deactivated.

(15) An exemplary embodiment of a fluid management system includes at least one fluid supply container, a pump, and a control system. The pump delivers fluid from the fluid supply container to a surgical site. The control system operates the pump and includes a user interface for communication between the control system and a user, a temporary adjustment device operatively connected to the pump, and at least one processor operatively connected to the user interface, the temporary adjustment device, and the pump. The user interface allows a user to input at least one of a setpoint flow rate and a setpoint pressure of the fluid being delivered from the pump. If the user inputs a setpoint flow rate, the temporary adjustment device allows a user to increase the setpoint flow rate of the fluid to an increased temporary flow rate and decrease the setpoint flow rate of the

fluid to a decreased temporary flow rate when the user activates the temporary adjustment device. If the user inputs a setpoint pressure, the temporary adjustment device also allows a user to increase the setpoint pressure of the fluid to an increased temporary pressure and decrease the setpoint pressure of the fluid to a decreased temporary pressure when the user activates the temporary adjustment device. The processor causes the pump to return to the setpoint flow rate and the setpoint pressure when the temporary adjustment device is deactivated.

(16) An exemplary embodiment of a fluid management system includes a control system having a printer and at least one processor operatively connected to the printer. The processor is configured to cause the printer to print one or more documents that display information relating to a surgical procedure at any time during the procedure, at set time increments during the procedure, and/or at the end of the procedure. The information relating to the surgical procedure includes at least one of a date of the surgical procedure, a type of the surgical procedure, a start time of the surgical procedure, an end time of the surgical procedure, a volume of fluid pumped during the surgical procedure, a fluid deficit of the surgical procedure, the fluid deficit at set time intervals during the surgical procedure, an average fluid pressure during the surgical procedure, a presence of fluid warming during the surgical procedure, an average fluid temperature during the surgical procedure, facility information, physician information, and patient information.

(17) An exemplary embodiment of a fluid management system for supplying at least two types of fluid to a surgical site includes a pump and a control system. The pump is configured to deliver a first fluid from a first fluid supply container and a second fluid from a second fluid supply container to the surgical site. The control system includes a user interface for communication between the control system and a user and at least one processor operatively connected to the user interface, the pump, and the first and second fluid supply containers. The processor monitors an amount of fluid in each of the first and second fluid supply containers to determine whether the first fluid or the second fluid is being delivered to the surgical site by the pump. The processor determines when one of the first and second fluids is being delivered by the pump to the surgical site and the other of the first and second fluids is desired at the surgical site by the user.

(18) A method of supplying at least two types of fluids to a surgical site includes configuring a control system of a fluid management system to instruct a user to place a first fluid supply container having a first fluid on a first hanging member of the fluid management system, and to instruct a user to place a second fluid supply container having a second fluid on a second hanging member of the fluid management system. The method further includes configuring the control system to instruct the user to fluidly connect the first fluid supply container to the pump after receiving instructions from the user to supply the first fluid to the surgical site, and to activate the pump to supply the first fluid to the surgical site. The method also includes configuring the control system to deactivate the pump after receiving instructions from the user to supply the second fluid type to the surgical site, and to instruct the user to fluidly disconnect the first fluid supply container from the pump and fluidly connect the second fluid type to type to the pump. The method further includes configuring the control system to activate the pump to supply the second fluid type to the surgical site.

(19) An exemplary embodiment of a fluid management system includes at least one fluid supply container, a pump for delivering fluid from the fluid supply container to a surgical instrument at the surgical site, and a control system for operating the pump. The control system includes a user interface for communication between the control system and a user and at least one processor operatively connected to the user interface and the pump. The control system is configurable to operate the fluid management system in one of a first pressure control mode, a second pressure control mode, and a third pressure control mode during a surgical procedure. The first pressure control mode includes the control system causing a first pressure of the fluid within the fluid management system to correspond to a first desired pressure within the fluid management system as set by the user via the user interface. The second pressure control mode includes the control

system causing a second pressure of the fluid at the surgical instrument to correspond to a second desired pressure of the fluid at the surgical instrument as set by the user via the user interface. A third pressure control mode includes the control system causing a third pressure of the fluid at a body cavity of a patient to correspond to a third desired pressure of the fluid at the body cavity as set by the user via the user interface.

(20) An exemplary method of enabling or disabling a heating function of a heating assembly for a fluid management system includes configuring the control system to detect a presence of a warming cartridge aligned with the heating assembly with one or more sensors of the fluid management system. The method further includes configuring the control system to enable the heating function of the heating assembly when the sensors detect the presence of the warming cartridge, and to disable the heating function when the sensors do not detect the presence of the warming cartridge.

(21) An exemplary embodiment of a fluid management system includes at least one fluid supply container, a pump for delivering fluid from the fluid supply container to the surgical site, and a control system for operating the pump. The control system includes a user interface for communication between the control system and a user and at least one processor operatively connected to the user interface and the pump. The control system operates the fluid management system in a first control mode that allows the user to adjust at least one of a visualization condition and a distention condition at the surgical site. The user adjusts the visualization condition by increasing or decreasing a visualization setting in the user interface, and the user adjusts the distention condition by increasing or decreasing a distention setting in the user interface. The processor makes adjustments to the speed of the pump based on the desired visualization condition and the desired distention condition as indicated by the user via the user interface.

(22) An exemplary embodiment of a fluid management system includes a main unit and one or more optional modules. The main unit has a pump configured to deliver fluid from at least one fluid supply container to a surgical site and a control system that includes at least one processor. The one or more optional modules are operatively connected to the processor of the control system. The one or more optional modules include at least one of a deficit module for monitoring an amount of fluid returning from the surgical site, an aspiration module for regulating a vacuum pressure applied to the surgical site, a fluid collection module for collecting fluid from the surgical site, a fluid suction and collection module for returning and collecting fluid from the surgical site, and a fluid flow and evacuation module. The fluid management system is configurable between a plurality of configurations. In a first configuration, none of the optional modules are operatively connected to the processor of the main unit. In a second configuration, one of the optional modules is operatively connected to the processor of the main unit. In a third configuration at least two of the optional modules are operatively connected to the processor of the main unit.

(23) An exemplary embodiment of a fluid management system includes a main unit, a deficit module, and a fluid collection module. The main unit has a pump for delivering fluid from at least one fluid supply container to a surgical site and a control system that includes at least one processor. The deficit module is removably operatively coupled to the main unit, and the deficit module is configured to monitor an amount of fluid returning from the surgical site. The fluid collection module is removably operatively coupled to the main unit, and the fluid collection module includes a collection chamber for receiving fluid from the surgical site.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an exemplary embodiment of a fluid management system for an operating room environment;

(2) FIG. 2 illustrates an exemplary embodiment of a main unit of the fluid management system of FIG. 1;

(3) FIG. 3 illustrates an exemplary embodiment of a heater and fluid conditioner assembly of the main unit of FIG. 2;

(4) FIG. 4 illustrates an exemplary cartridge assembly for insertion into the heater and fluid conditioner assembly of FIG. 3;

(5) FIG. 5 illustrates an exemplary fluid conditioner and fluid warming cartridge of the cartridge assembly of FIG. 4 disconnected from each other;

(6) FIG. 6 illustrates an exemplary fluid path through the cartridge assembly of FIG. 4;

(7) FIG. 7 illustrates an exploded view of an IR lamp sub-assembly of the heater assembly of FIG. 3 with the cartridge assembly of FIG. 4;

(8) FIG. 7A illustrates an exemplary embodiment of a threshold detector for a crossover circuit;

(9) FIG. 7B illustrates an exemplary embodiment of a relay bank for interacting with the crossover circuit of FIG. 7A, where the switches of the relay bank are in a first position;

(10) FIG. 7C illustrates the relay bank of FIG. 7B, where the switches of the relay bank are in a second position;

(11) FIG. 8 illustrates another exemplary embodiment of a fluid conditioner for insertion into a fluid conditioning assembly of a fluid management system;

(12) FIG. 9 illustrates an exemplary alignment between the fluid conditioner of FIG. 5 and the fluid conditioning assembly of FIG. 3;

(13) FIG. 10 illustrates a cross-sectional view of an exemplary embodiment of a fluid conditioner for the cartridge assembly of FIG. 4;

(14) FIG. 11 illustrates the fluid conditioner of FIG. 10 showing locations of the fluid conditioner that are aligned with sensors of the fluid conditioning assembly of FIG. 3;

(15) FIG. 12 illustrates a perspective view of the fluid conditioner of FIG. 10;

(16) FIG. 13 illustrates an exploded perspective view of the fluid conditioner of FIG. 10;

(17) FIG. 14 illustrates a perspective view of an exemplary fluid warming cartridge of the cartridge assembly of FIG. 4;

(18) FIG. 15 illustrates an exploded perspective view of the fluid warming cartridge of FIG. 14;

(19) FIG. 16 illustrates a front view of the fluid warming cartridge of FIG. 14 when fluid moving through it is at a low pressure;

(20) FIG. 17 illustrates a front view of the fluid warming cartridge of FIG. 14 when fluid moving through it is at a high pressure;

(21) FIG. 18 illustrates a cross-sectional front view of the fluid warming cartridge of FIG. 14 when fluid moving through it is at a high pressure;

(22) FIG. 19 illustrates an exemplary embodiment of a main unit for the fluid management system of FIG. 1, where the main unit includes an opening to draw air through the heater assembly of the main unit for cooling purposes and another opening to exhaust the resulting warm air onto or near one or more of the fluid supply bags or containers that are hanging from the fluid management system in order to pre-warm the fluid;

(23) FIG. 20 illustrates a perspective view of an exemplary embodiment of a deficit module and a deficit cartridge for the fluid management system of FIG. 1, where the deficit cartridge is inserted in the deficit module;

(24) FIG. 21 illustrates a front view of the deficit module and deficit cartridge of FIG. 20;

(25) FIG. 22 illustrates a right-side perspective view of the deficit module and deficit cartridge of FIG. 20, where the deficit cartridge is removed from the deficit module;

(26) FIG. 23 illustrates a left side perspective view of the deficit module and the deficit cartridge of FIG. 20, where the deficit cartridge is removed from the deficit module;

(27) FIG. 24 illustrates a rear view of the deficit cartridge of FIG. 20;

(28) FIG. 25 illustrates a side view of the deficit cartridge of FIG. 20;

(29) FIG. 26 illustrates a side view of the deficit cartridge of FIG. 20 with valve and port covers removed;

(30) FIG. 27 illustrates the deficit cartridge of FIG. 20 aligned with a moveable manifold of the deficit module of FIG. 20, wherein the movable manifold is in an open position relative to the deficit cartridge;

(31) FIG. 28 illustrates the deficit cartridge of FIG. 20 aligned with a moveable manifold of the deficit module of FIG. 20, where the movable manifold is in a closed position relative to the deficit cartridge such that connections are made between the deficit cartridge and the deficit module;

(32) FIGS. 28A-28C illustrate an exemplary connection between a port of the deficit cartridge of FIG. 20 and a connector of the movable manifold of the deficit module of FIG. 20;

(33) FIG. 29 illustrates an exploded perspective view of the deficit cartridge of FIG. 20;

(34) FIG. 30 illustrates a front view of the deficit cartridge of FIG. 20 aligned with non-contact fluid presence sensors of the deficit module of FIG. 20;

(35) FIG. 31 illustrates a perspective view of the deficit cartridge of FIG. 20 aligned with non-contact fluid presence sensors of the deficit module of FIG. 20;

(36) FIG. 32 illustrates a side view of the deficit cartridge of FIG. 20;

(37) FIG. 33 illustrates a side view of the deficit cartridge of FIG. 20 when the deficit monitoring feature of the fluid management system is in a fill/measure cycle;

(38) FIG. 34 illustrates a side view of the deficit cartridge of FIG. 20 when the deficit monitoring feature of the fluid management system is in a fill/evacuation cycle;

(39) FIG. 35 illustrates a cross-sectional top view of the deficit module of FIG. 20;

(40) FIG. 36 illustrates an exploded perspective view of the deficit module of FIG. 20;

(41) FIG. 37 illustrates a perspective view of an exemplary deficit pump manifold assembly for the deficit module of FIG. 20;

(42) FIG. 38 illustrates an exploded perspective view of an exemplary deficit cartridge receiving assembly of the deficit module of FIG. 20 for receiving the deficit cartridge of FIG. 20;

(43) FIG. 39 illustrates a perspective view of an exemplary manifold connection assembly of the deficit module of FIG. 20 and the deficit cartridge receiving assembly of FIG. 38;

(44) FIG. 40 illustrates a left side perspective view of the deficit module of FIG. 20;

(45) FIG. 41 illustrates a right-side perspective view of the deficit module of FIG. 20;

(46) FIG. 42 illustrates a top view of the deficit module of FIG. 20;

(47) FIG. 43 illustrates a cross-sectional view of the deficit module of FIG. 20 taken along the line A-A shown in FIG. 42;

(48) FIG. 44 illustrates a partial view of the deficit module shown in FIG. 43 showing an exemplary manifold connection assembly for connecting the deficit cartridge shown in FIG. 20, where the manifold connection assembly is in a disengaged position relative to the deficit cartridge;

(49) FIG. 45 illustrates the exemplary engagement mechanism shown in FIG. 44, where the manifold connection assembly is in the disengaged position relative to the deficit cartridge;

(50) FIG. 46 illustrates a top view of the deficit module of FIG. 20;

(51) FIG. 47 illustrates a cross-sectional view of the deficit module of FIG. 20 taken along the line B-B shown in FIG. 46;

(52) FIG. 48 illustrates a partial view of the deficit module shown in FIG. 43 showing the exemplary manifold connection assembly of FIG. 44, where the manifold connection assembly is in an engaged and connected position relative to the deficit cartridge;

(53) FIG. 49 illustrates the exemplary engagement mechanism shown in FIG. 44, where the manifold connection assembly is in the engaged and connected position relative to the deficit cartridge;

(54) FIG. 50 illustrates another exemplary embodiment of the fluid management system shown in FIG. 1;

(55) FIG. 51 illustrates an exemplary embodiment of a fluid flow monitoring and evacuation

module for the fluid management system of FIG. 50;

(56) FIG. 52 illustrates a perspective view of an exemplary embodiment of an aspiration module and a pressure regulator for the fluid management system of FIG. 1, where the pressure regulator is inserted in the aspiration module;

(57) FIG. 53 illustrates a schematic view of an exemplary embodiment of the pressure regulator shown in FIG. 52;

(58) FIG. 54 illustrates a schematic view of another exemplary embodiment of the pressure regulator shown in FIG. 52;

(59) FIG. 55 illustrates a schematic view of the pressure regulator shown in FIG. 54 showing valves of the pressure regulator disposed in a series;

(60) FIG. 56 illustrates a rear perspective view of an exemplary embodiment of the pressure regulator shown in FIG. 54;

(61) FIG. 57 illustrates a front perspective view of the pressure regulator shown in FIG. 56;

(62) FIG. 58 illustrates an exploded perspective view of the pressure regulator shown in FIG. 56;

(63) FIG. 59 illustrates a top view of the pressure regulator shown in FIG. 56;

(64) FIG. 60 illustrates a rear view of the pressure regulator shown in FIG. 56;

(65) FIG. 61 illustrates a bottom view of the pressure regulator shown in FIG. 56;

(66) FIG. 62 illustrates a side view of the pressure regulator shown in FIG. 56;

(67) FIG. 63 illustrates a cross-sectional view of the pressure regulator shown in FIG. 56 taken along the lines C-C shown in FIG. 61;

(68) FIG. 64 illustrates a perspective view of an exemplary connection between the pressure regulator shown in FIG. 56 and an exemplary receiving mechanism for the aspiration module shown in FIG. 52;

(69) FIG. 65 illustrates a perspective view of another exemplary embodiment of the pressure regulator shown in FIG. 54;

(70) FIG. 66 illustrates an exploded perspective view of the pressure regulator shown in FIG. 65;

(71) FIG. 67 illustrates a partial perspective view of the pressure regulator shown in FIG. 65;

(72) FIG. 68 illustrates a top view of the pressure regulator shown in FIG. 65;

(73) FIG. 69 illustrates a cross-sectional view of the pressure regulator shown in FIG. 65 taken along the lines D-D shown in FIG. 68;

(74) FIG. 70 illustrates a cross-sectional view of the pressure regulator shown in FIG. 65 taken along the lines E-E shown in FIG. 68;

(75) FIG. 71 illustrates a cross-sectional view of the pressure regulator shown in FIG. 65 taken along the lines F-F shown in FIG. 68;

(76) FIG. 72 illustrates a top cross-sectional view of an exemplary connection between the pressure regulator shown in FIG. 65 and an exemplary receiving mechanism for the aspiration module shown in FIG. 52;

(77) FIG. 73 illustrates a partial view of the exemplary connection between the pressure regulator and receiving mechanism shown in FIG. 72 showing a connection between a port of the pressure regulator and a port of the aspiration module;

(78) FIG. 74 illustrates a side cross-sectional view of an exemplary embodiment of the aspiration module shown in FIG. 52;

(79) FIG. 75 illustrates a rear perspective view of the aspiration module shown in FIG. 74;

(80) FIG. 76 illustrates an exemplary prompt by the fluid management system of FIG. 1 to a user via a user interface regarding the type of procedure to be performed;

(81) FIG. 77 illustrates another exemplary prompt by the fluid management system of FIG. 1 to a user via a user interface regarding the type of procedure to be performed;

(82) FIG. 78 illustrates an exemplary prompt by the fluid management system of FIG. 1 to a user via a user interface regarding the number of fluid types that will be used during a procedure;

(83) FIG. 79 illustrates another exemplary prompt by the fluid management system of FIG. 1 to a

user via a user interface regarding the number of fluid types that will be used during a procedure;

(84) FIG. **80** illustrates another exemplary prompt by the fluid management system of FIG. **1** to a user via a user interface regarding the number of fluid types that will be used during a procedure;

(85) FIG. **81** illustrates an exemplary embodiment of a procedure run screen on a user interface of a fluid management system when the fluid management system is set to a pressure control mode;

(86) FIG. **81A** illustrates a means of calculating a compensation height for calculating a pressure at a surgical scope or instrument;

(87) FIG. **82** illustrates an exemplary embodiment of a procedure settings screen on a user interface of a fluid management system;

(88) FIG. **83** illustrates an exemplary embodiment of a procedure run screen on a user interface of a fluid management system when the fluid management system is set to a “Surgical Site” control mode;

(89) FIG. **84** illustrates a flow chart for a fluid management system that is operating in the surgical site control mode shown in FIG. **83**;

(90) FIG. **85** illustrates an exemplary embodiment of a procedure settings screen on a user interface of a fluid management system that includes operating a bolus device;

(91) FIG. **86** illustrates the procedure settings screen of FIG. **85**;

(92) FIG. **87** illustrates the procedure settings screen of FIG. **85**;

(93) FIG. **88** illustrates an exemplary embodiment of a procedure settings screen on a user interface of a fluid management system that includes operating a printer of the fluid management system;

(94) FIG. **89** illustrates an exemplary embodiment of a procedure settings screen on a user interface of a fluid management system that includes alert settings for notifying a user when a fluid supply container is becoming depleted, where the system is in a timed setting;

(95) FIG. **90** illustrates the procedure settings screen of FIG. **89**, wherein the system is in a percentage setting;

(96) FIG. **91** illustrates the procedure settings screen of FIG. **89**, wherein the system is in a volume setting;

(97) FIG. **92** illustrates an exemplary embodiment of a fluid management system for a physician's office environment, where fluid bags are attached to hanging members of the fluid management system; and

(98) FIG. **93** illustrates the fluid management system of FIG. **92**, where a fluid supply bag and a fluid return canister are attached to hanging members of the fluid management system.

DETAILED DESCRIPTION

(99) The Detailed Description describes exemplary embodiments of the invention and is not intended to limit the scope of the claims in any way. Indeed, the invention is broader than and unlimited by the exemplary embodiments, and the terms used in the claims have their full ordinary meaning, unless otherwise noted in the application. Features and components of one exemplary embodiment may be incorporated into the other exemplary embodiments. Inventions within the scope of this application may include additional features, or may have less features, than those shown in the exemplary embodiments.

(100) As described herein, when one or more components are described as being connected, joined, affixed, coupled, attached, or otherwise interconnected, such interconnection may be direct as between the components or may be indirect such as through the use of one or more intermediary components. Also as described herein, reference to a “member,” “component,” or “portion” shall not be limited to a single structural member, component, or element but can include an assembly of components, members, or elements. Also as described herein, the terms “substantially” and “about” are defined as at least close to (and includes) a given value or state (preferably within 10% of, more preferably within 1% of, and most preferably within 0.1% of).

(101) In endoscopic surgical procedures, steady distention and clear visibility are important to procedural efficacy and efficiency. Fluid management systems are used to provide fluid to a

surgical site such that a surgeon has the desired distention and visualization while performing a surgical procedure. Fluid management systems can also be used to remove fluid from the surgical site. The various embodiments of fluid management systems described herein relate to modular systems that include software-controlled, electro-mechanical devices or modules that may be used in combination with single or multiuse tubing sets. The modular, surgical fluid management systems described herein are fully configurable to meet user needs based on, for example, the types of surgical procedures being performed and the surgical environment. Exemplary functions of the fluid management systems described herein include fluid pressurization, fluid warming, fluid deficit monitoring, suction, suction regulation, fluid collection, and/or fluid evacuation into a facility's waste disposal system. The fluid management systems can be configured based on surgical discipline (e.g., gynecological, urological, and/or orthopedic procedures) and environment (e.g., operating room or physician's office), as well as based on other needs and/or preferences of the user and/or facility. The fluid management systems may be capable of integrated suction and fluid collection and/or may be compatible with third-party suction and fluid collection devices, as well as central suction systems of facilities where the fluid management systems are used.

(102) Referring to FIG. 1, an exemplary embodiment of a fluid management system **100** for an operating room environment where gynecological, urological, and orthopedic procedures are performed is shown. The system **100** includes an elevated structure **101**, a main unit **102**, a deficit module **104**, a fluid collection module **106**, and a fluid evacuation module **108**. The system **100** may also include an aspiration module **5201** (FIG. 52), and/or a fluid flow and evacuation module **5101** (FIG. 51). In some embodiments, the elevated structure **101** includes wheels **103** such that the system **100** can be moved to a desired location within the operating room or to a storage area. The system **100** may be modular, such that the system **100** described above can be configured as desired by the user.

(103) The main unit **102** may have a control system that includes one or more processors (not shown) for controlling and/or communicating with the various modules and components of the system **100** or other facility equipment. The various modules and components may also have one or more processors (not shown) for performing designated functions and/or communicating with the control system of main unit **102** or other facility equipment. The processor(s) may execute instructions (e.g., software code) stored in memory (not shown) of the system **100** and/or execute instructions inputted into the system by a user. In some embodiments, the control system may have “Bluetooth” capability for connecting to remotely located components or modules of the system **100** or other facility equipment and “Wi-Fi” capability for connecting to the internet. The control system may include a touch-screen graphical user interface **110** for receiving one or more inputs from a user and displaying information of the system **100** (e.g., information regarding fluid pressure, fluid volume, fluid temperature, fluid deficit, etc.).

(104) Referring to FIGS. 1 through 3, the main unit **102** may also include a pump **212** (e.g., a peristaltic pump) for fluid pressurization, a heater assembly **314** for fluid warming, a fluid conditioning assembly **315** for sensing one or more fluid characteristics (e.g., fluid presence, temperature, etc.), hanging members **116** (e.g., hooks) for hanging fluid supply and/or return containers (e.g., bags, canisters, vessels, etc.), and a printer **218** for printing out pertinent procedure information (e.g., information regarding procedure type, procedure start time, procedure end time, total fluid volume, average fluid pressure, total fluid deficit, deficit by fluid type, average fluid temperature, etc.) during or after the surgical procedure. The processor of the control system can be in communication with the pump **212**, heater assembly **314**, fluid conditioning assembly **315**, pressure sensors **949** (FIG. 9), solenoid valve **951** (FIG. 9), hanging members **116**, printer **218**, deficit module **104**, fluid collection module **106**, fluid evacuation module **108**, aspiration module **5201** (FIG. 52), fluid flow and evacuation module **5101** (FIG. 51), and/or any other component of the system **100**.

(105) The pump **212** may be fluidly connected to the fluid container(s) that are hanging on the

hanging members **116** such that the pump can pump fluid through a tubing set to a surgical scope or instrument (e.g., hysteroscope, cystoscope, ureterscope, nephroscope, etc.) at a surgical site. The tubing set may include a fluid conditioner (e.g., fluid conditioner **420** shown in FIG. **4** and described in the present application) that works in combination with one or more non-contact sensors (e.g., non-contact sensors of the fluid conditioning assembly **315** or any other non-contact sensors in the system **100**) such that the system **100** can monitor one or more characteristics of the fluid that is moving to the surgical site. The tubing set may also include a fluid warming cartridge (e.g., fluid warming cartridge **422** shown in FIG. **4** and described in the present application) that works in combination with the heater assembly **314** such that the system **100** can warm fluid that is moving to the surgical site.

(106) A suction source pulls fluid from the surgical site, through a tubing set, and either into a collection container of the collection module **106**, into a third-party fluid collection system, or into the waste disposal system of the facility in which the system **100** is being used. In certain embodiments, the suction source is a vacuum pump that is integral to main unit **102** or the fluid collection module **106**. In some embodiments, fluid collection module also includes a pump and one or more filters such that the fluid collection module can evacuate and filter surgical smoke to eliminate potentially hazardous byproducts of electrosurgical procedures.

(107) Referring to FIG. **1**, the fluid collection module **106** may be independently mobile and removably coupled to the elevated structure **101** such that the module **106** can be removed from the elevated structure **101** and transported to a waste disposal area or room for disposal of the collected fluid. In some embodiments, the collection container of the collection module **106** may include disposable liners that can easily be replaced after the fluid has been evacuated from the suction and collection module **106** and into the facility's waste disposal system. In some embodiments, the suction source is external to the system **100** and pulls fluid to either the collection container of the collection module **106**, a third party-fluid collection system, or the waste disposal system of the facility. In embodiments in which the fluid is pulled directly into the waste disposal system of the facility, the collection module **106** may be bypassed or removed from the system **100** during use (e.g., as shown in FIG. **50**). The fluid collection module **106** may include a processor that communicates with the main unit **102**, the deficit module **104**, the aspiration module **5201** (FIG. **52**), other components of the system **100**, and/or other facility equipment. In some embodiments, the fluid collection module **106** may include a weight measuring mechanism (e.g., a scale) that allows the fluid management system **100** to determine a volume of fluid returning from the surgical site for fluid outflow and/or deficit monitoring purposes.

(108) Prior to the fluid moving into the collection module **106**, a third-party fluid collection system, or a waste disposal system of the facility, the fluid may move through a single or multiuse deficit cartridge (e.g., deficit cartridge **2010** shown in FIG. **25** and described in the present application) such that the system **100** can calculate and monitor a fluid deficit between fluid being provided to the surgical site and fluid being returned from the surgical site. The deficit cartridge may work in combination with the deficit module **104** (or the fluid flow and evacuation module **5101** shown in FIG. **51** and described in the present application) and the main unit **102** to allow the system **100** to calculate and monitor the fluid deficit.

(109) In certain embodiments, system **100** includes an aspiration module (e.g., aspiration module **5201** shown in FIG. **52** and described in the present application), and a single or multiuse pressure regulator (e.g., the pressure regulators **5205** shown in FIGS. **52-73** and described in the present application) that is fluidly connected to the tubing set and the suction source. The pressure regulator and aspiration module may work in combination with each other and the main unit **102** to regulate a vacuum pressure provided to the surgical site by the suction source to pull fluid from the surgical site.

(110) FIGS. **4** through **6** illustrate an exemplary embodiment of a cartridge assembly **419** for a single or multiuse disposable tubing set of the system **100**, where the cartridge assembly **419**

includes a fluid conditioner **420** and a fluid warming cartridge **422**. The fluid conditioner **420** is configured to connect to the warming cartridge **422** to form the cartridge assembly **419** (as shown in FIG. 4). For example, referring to FIG. 5, the fluid conditioner **420** may have one or more connection members **421** that are configured to connect to one or more connection members **423** of the fluid warming cartridge **422**. The connection members **421**, **423** of the fluid conditioner **420** and the fluid warming cartridge **422** may be connected by, for example, a snap-fit connection, a friction fit connection, etc. In other embodiments, the fluid conditioner **420** and the fluid warming cartridge **422** may be connected by gluing, ultrasonically welding, or any other suitable means of joining the fluid conditioner and the fluid warming cartridge. In certain embodiments, the cartridge assembly **419** is a single, fully integrated component with combined fluid conditioning and fluid warming functions. In these embodiments, the single, fully integrated component of the cartridge assembly **419** can be, for example, a single injection molded component. In certain embodiments, the cartridge assembly **419** is provided as a fully assembled component of a single or multiuse tubing set. In some embodiments, the fluid conditioner **420** is provided as a fully assembled component of a single or multiuse tubing set (e.g., including the fluid conditioner **420** and tube **841** assembly shown in FIG. 8), and the warming cartridge **422** is provided as an accessory component that can be attached to the fluid conditioner **420** if desired. In such embodiments, the user may configure the tubing set for fluid warming by removing tube **841** (FIG. 8) from the fluid conditioner **420** and connecting the warming cartridge **422** to the fluid conditioner **420**.

(111) In certain embodiments, the main unit **102** can sense whether the fluid conditioner **420** has been inserted alone (e.g., without the warming cartridge **422**) into the system **100** or the cartridge assembly **419** (that includes the fluid conditioner **420** and warming cartridge **422**) has been inserted into the system. For example, the main unit **102** may include one or more sensors (e.g., proximity sensors, mechanical sensors, optical sensors, laser sensors, etc.) that can detect whether the fluid conditioner **420** alone or the cartridge assembly **419** was inserted into the system **100**. The control system of the system **100** can then enable the fluid warming function of the system **100** (e.g., the heater assembly **314** shown in FIG. 3) when a warming cartridge **422** is inserted into the system **100** and disable the warming function when a warming cartridge **422** is not inserted into the system **100**.

(112) Referring to FIG. 6, during use of the system **100**, fluid may be pumped through a first tube **624** of the tubing set and into an inlet port **625** of the fluid conditioner **420**. The fluid then flows along a first flow path **626** through an inlet chamber **1053** (FIG. 10) of the fluid conditioner, moves through an outlet port **527** (FIG. 5) of the fluid conditioner **420**, and through an inlet opening **528** (FIG. 5) of the fluid warming cartridge **422**. The fluid then moves along a first side **1671** (FIGS. 16-18) of the warming cartridge **422** along a fluid path **629**, moves through a connector or tube **530**, and into a second side **1670** (FIGS. 16-18) of the fluid warming cartridge **422** along a path **631**. Subsequently, the fluid exits an outlet opening **532** (FIG. 5) and moves through inlet port **533** (FIG. 5) of an outlet chamber **1054** (FIG. 10) of the fluid conditioner **420**, where the fluid moves along a path **636** such that the fluid exits outlet **634** of the fluid conditioner **422** and moves through a tube **635** of the disposable tubing set to a surgical instrument at a surgical site. The connector or tube **530** is shown as having a U-shape, but the connector or tube can take any suitable form that causes the first and second sides of the warming cartridge **422** to be fluidly connected. While the first and second sides of the fluid warming cartridge are shown being fluidly connected by the connector or tube **530**, it should be understood that the first and second sides can be fluidly connected without the need for the connector or tube **530**. For example, the warming cartridge **422** can have a channel that fluidly connects the first and second sides.

(113) In the illustrated embodiment, the fluid enters the fluid path **629** through the inlet opening **528** (FIG. 5) of the warming cartridge **422** at a lower position relative to the exit of the fluid path **629** at the inlet of the connector or tube **530**, and the fluid enters the fluid path **631** at the exit of the connector or tube **530** at a lower position relative to the outlet opening **532** (FIG. 5) of the warming

cartridge **422**. The enter low, exit high configuration for each of the fluid paths **629**, **631** promotes a more uniform, controlled warming by reducing Eddy currents and areas of stagnant flow. While the fluid is shown taking the fluid paths **629**, **631** through the warming cartridge **422**, it should be understood that the fluid can take any suitable path through the warming cartridge **422**.

(114) Inserting the cartridge assembly **419** into the main unit **102** of the system **100** aligns the fluid conditioner **420** with the fluid conditioning assembly **315** (FIG. 3) and the fluid warming cartridge **422** with the heater assembly **314** (FIG. 3). The fluid conditioner **420** may have a handle **442** that allows a user to easily insert the cartridge assembly **419** into the main unit **102**.

(115) Referring to FIG. 7, the heater assembly **314** (FIG. 3) may include an IR lamp assembly **737** used to warm the fluid moving along the fluid paths **629**, **631** (FIG. 6) of the warming cartridge **422**. The IR lamp assembly **737** may include a support structure **738**, one or more elongated IR lamps with IR reflective coatings **739** disposed on each side of the warming cartridge **422**, and a parabolic reflector **740** disposed on each side of the warming cartridge **422** such that the parabolic reflector **740** focuses the IR energy on the fluid paths. The heater assembly **314** may, however, utilize other types of IR lamps such as bulbs, rings, panels, circular modules, or any other suitable forms that are capable of warming fluid moving through the warming cartridge **422** or any other cartridge, tube, or vessel capable of exposing the fluid to IR lamps.

(116) Referring to FIG. 8, in some embodiments, fluid warming may not be desired or necessary during a procedure. In these embodiments where fluid warming cartridge **422** is not necessary, a connector or tube **841** is used to connect the inlet chamber **1053** (FIG. 10) and the outlet chamber **1054** (FIG. 10) of the fluid conditioner **420**. While the inlet and outlet chambers are shown being fluidly connected by the connector or tube **841**, it should be understood that the inlet and outlet chambers can be fluidly connected without the need for the connector or tube **841**. For example, the fluid conditioner **420** can have a channel that fluidly connects the inlet and outlet chambers.

(117) In an alternative embodiment, rather than utilizing the connector or tube **841**, the fluid conditioner **420** may be included in a cartridge assembly that has a pulse damping component (not shown) that is similar in construction to the warming cartridge **422** described below with reference to FIGS. 14-18, but the fluid damping component is not used for fluid warming. For example, the pulse damping component may include a rigid body (e.g., similar to rigid body **1472** shown in FIGS. 14-18) and flexible side sheets (e.g., similar to flexible side sheets **1473**, **1474** shown in FIGS. 14-18), where the rigid body and flexible side sheets at least partially define a fluid path that connects the inlet chamber **1053** (FIG. 10) of the fluid conditioner **420** to the outlet chamber **1054** (FIG. 10) of the fluid conditioner **420**. In alternative embodiments, the pulse damping component may comprise a flexible vessel or channel without a rigid body, in which the flexible vessel or channel defines a fluid path that fluidly connects to the inlet chamber **1053** (FIG. 10) and the outlet chamber **1054** (FIG. 10) of the fluid conditioner **420**. In any of the embodiments described above, the flexible vessel or channel is capable of expanding and contracting to dampen the fluid pulsations. That is, the flexible vessel or flexible side sheets can expand and contract to reduce pulsations of the fluid as pressure of the fluid moving through the conduit fluctuates. This damping of the fluid pulsations facilitates steady distention and good visualization during a surgical procedure. The fluid conditioner **420** and pulse damping component can be connected by any suitable means, such as, for example, any means discussed in the present application regarding the connection of the fluid conditioner **420** and the fluid warming cartridge **422**. In certain embodiments, the fluid conditioner and pulse damping component can be included in an integrated cartridge assembly where the fluid conditioner **420** and fluid damping component are included in a single cartridge. In certain embodiments, the pulse damping component with a rigid body and flexible side sheets or the flexible vessel or channel used for pulse damping may not be connected to fluid conditioner **420**, but instead be connected in the tubing set between the outlet port **634** (FIG. 10) and the surgical site.

(118) Referring to FIG. 9, the fluid conditioner **420** is configured to connect or align with one or

more non-contact sensors (e.g., sensors **943-950**) of the fluid conditioning assembly **315** such that the sensors can sense one or more characteristics of the fluid without contacting the fluid. For example, the fluid conditioning assembly **315** may include one or more fluid presence sensors (**943, 947, 948, 950**), one or more fluid temperature sensors (**944, 945, 946**), and a port **1062** (FIG. **10**) that connects to one or more pressure sensors **949** located in the main unit **102**. The port **1062** (FIG. **10**) that connects to one or more pressure sensors **949** may also connect to a solenoid valve **951** for expelling excess air that has accumulated in the fluid conditioner **420**. The pressure sensors **949** and the solenoid valve **951** may be connected to the port **1062** by one or more tubes or conduits and connection component **952**. The control system of the fluid management system **100** may be configured to at least partially control the pressurization of the fluid by pump **212**, the warming of fluid by the heater assembly **314**, and the expelling of air from the fluid conditioner **420** based on the interface between the fluid conditioning assembly **315** (FIG. **3**) and the fluid conditioner **420**.

(119) Referring to FIGS. **10** through **13**, an exemplary embodiment of the fluid conditioner **420** may include a rigid body **1052** that defines a first or inlet chamber **1053** and a second or outlet chamber **1054**. In some embodiments, fluid conditioner **420** may include a fully or partially enclosed middle chamber **1075** located between the inlet chamber **1053** and the outlet chamber **1054** to provide a separation gap between walls of the inlet and outlet chambers. This separation gap created by the middle chamber **1075** prevents heat transfer between incoming and outgoing fluid that would occur if the inlet chamber **1053** shared a common wall with the outlet chamber **1054**. The rigid body **1052** can be, for example, an injection molded body. Referring to FIGS. **12** and **13**, the fluid conditioner **420** may also include a film **1255** that is connected to the rigid body **1052** to further define and enclose the chambers **1053, 1054** to create flow paths. The film **1255** can be connected to the rigid body **1052** by gluing, laser welding, ultrasonic welding, or any other suitable means. The film **1255** is configured to allow one or more sensors of the sensing assembly **315** (FIG. **3**) to sense one or more characteristics of the fluid moving through the inlet and outlet chambers **1053, 1054** without contacting the fluid. The film **1255** can be, for example, a plastic film. In alternative embodiments, the fluid conditioner **420** does not include the film **1255**, but rather the fluid conditioner **420** is a rigid vessel that is configured to allow one or more sensors of the sensing assembly **315** (FIG. **3**) to sense one or more characteristics of the fluid without contacting the fluid. In some of these embodiments, a portion of the rigid vessel that aligns with the sensors of the sensing assembly can have a reduced thickness relative to the remainder of the fluid vessel that allows the sensors to sense characteristics of the fluid. In the embodiments mentioned above, the inlet chamber **1053** may have an inlet port **625** and an outlet port **527**, and the outlet chamber **1054** may have an inlet port **533** and an outlet port **634**. Outlet port **527** and inlet port **533** can have O-rings (e.g., O-rings **1363** shown in FIG. **13**) for making water-tight connections. In certain embodiments, the inlet port **625** and outlet port **634** can have barbed and/or glued portions for connecting to fluid tubing.

(120) Referring to FIGS. **9** through **11**, the fluid inlet chamber **1053** is aligned with a fluid presence sensor **943** that targets area **1156** and a fluid inlet temperature sensor **944** that targets area **1157**. Operation of the pump **212** (FIG. **2**) causes fluid to flow from a fluid supply bag or container through inlet port **625** into the inlet chamber **1053**. The inlet chamber **1053** may have a protruding wall **1058** that causes a section of the chamber to become thin or shallow, which mitigates air bubble stagnation by causing laminar flow through this section. The fluid presence sensor **943** verifies that fluid is present in the inlet chamber **1053** and, therefore, can be used by the system **100** to monitor performance and identify any problems. For example, if the pump is operating, but fluid presence sensor **943** is not detecting fluid, the control system may notify the user to check for a disconnected tubing line or possible occlusions of the fluid path between the fluid containers and the fluid conditioner **420**, such as, for example, kinked tubing or closed clamps.

(121) The fluid temperature sensor **944** may have several functions. For example, in embodiments

in which the heater assembly **314** is used to warm the fluid to a desired temperature (e.g., a temperature inputted by the user or a default system temperature), fluid temperature sensor **944** allows the control system to monitor the temperature of the fluid entering the warming cartridge **422** such that the control system can adjust the amount of IR energy provided by the heater assembly **314** to cause the fluid entering the outlet chamber **1054** of the fluid conditioner **420** to be at the desired temperature. In addition, if the user has hung pre-warmed fluid bags with fluid temperatures at high, potentially unsafe levels, the control system may disable the pump **212** and/or the heater assembly **314**, and then notify the user that such operations will remain disabled until the fluid temperature has sufficiently cooled or the fluid supply bags or containers have been changed. Alternatively, the control system may continue operation while increasing air flow through the heater assembly **314** to sufficiently cool the fluid before it reaches the outlet chamber **1054** of the fluid conditioner **420**. If such attempt fails, the fluid outlet temperature sensor **945** that targets area **1159** and/or fluid high-limit or thermal cut-off temperature sensor (“TCO Sensor”) **946** that targets area **1160** will cause the control system to disable the fluid pumping and warming operations until the temperature of the fluid has sufficiently cooled. Additionally, assuming an operating room environment where the user has enabled the fluid warming function, the temperature sensor **944** can be used to notify the user if the temperature of the fluid entering the fluid conditioner **420** may be too cool to achieve the desired fluid temperature. Finally, the control system can also determine if there is a problem with heater assembly **314**. For example, if the temperature sensor **944** detects that the temperature entering inlet chamber **1053** is acceptable, but sensor **945** detects that the temperature of the fluid did not achieve the desired fluid temperature, the control system will notify the user that there may be a problem with the heater assembly **314**.

(122) Still referring to FIGS. **9** through **11**, the outlet chamber **1054** of the fluid conditioner **420** may be designed to separate air bubbles from the fluid being delivered to the surgical site that may have been caused by fluid bag changes or the fluid warming process. For example, the fluid outlet chamber **1054** may have a substantially vertical wall or baffle **1061** (FIGS. **10-11**) that causes air bubbles to separate from the fluid when the fluid engages the wall. As shown in the illustrated embodiment, the baffle **1061** may not be connected to a perimeter of the outlet chamber **1054**.

(123) In certain embodiments, the outlet chamber **1054** is designed to facilitate fluid pressure monitoring and control via pressure sensors **949** located in the main unit **102**. For example, insertion of the fluid conditioner **420** may cause a connection between the outlet chamber **1054** of the fluid conditioner **420** with pressure sensors **949** located in main unit **102** via pressure port **1062** and one or more tubes or conduits (not shown). As the pressure of the pocket of air trapped between the fluid in outlet chamber **1054** and the pressure sensors **949** is indicative of the fluid pressure, the control system monitors the fluid pressure being read by the pressure sensors **949** in relation to the setpoint fluid pressure, and the control system adjusts the speed of the pump **212** to achieve and maintain the setpoint fluid pressure. To ensure pressure monitoring accuracy and guard against over pressure conditions, the control system constantly compares the readings of the pressure sensors **949** to ensure they are the same excepting normal tolerances for such sensors. Independent of software, the control system may employ redundant hardware circuits that disable or reverse the pump **212** if the fluid pressure exceeds the maximum allowable pressure for the procedure.

(124) To ensure that the pressure sensors **949** remain isolated from the fluid, the outlet chamber **1054** is designed not only to maintain a pocket of air between the pressure sensors **949** and the fluid, but also to include a hydrophobic filter **1065** that acts as a fluid barrier. Such hydrophobic filter **1065** may also act as a bacterial barrier to preserve the sterility of the fluid. To protect the hydrophobic filter **1065** from coming into contact with fluid entering the outlet chamber **1054** under turbulent or high flow conditions, outlet chamber **1054** may include an arcing wall or barrier **1066** that, in combination with the baffle **1061**, ensures that any fluid going over the top of the baffle **1061** is directed away from the hydrophobic filter. The pressure port **1062** may also include

an O-ring **1364** (FIG. **13**) for making a fluid tight connection.

(125) In addition to the presence sensor **943** for the inlet chamber **1053**, there may be at least three additional fluid presence sensors (**947**, **948**, **950**) that are aligned with the outlet chamber **1054**. The fluid presence sensor **947** (“fluid outlet sensor”) located at the outlet port **634** of the outlet chamber **1054** targets area **1167** and is used to ensure proper flow of fluid through the fluid conditioner **420**. For example, if the control system detects that the pump **212** is pumping fluid, but the fluid outlet sensor **947** is not detecting fluid, the control system may disable the pump **212** and/or notify the user of a problem with the system **100**. In addition, if the fluid warming function is present and enabled, the fluid outlet sensor **947** ensures that the fluid warming cartridge **422** is full of fluid before the fluid warming function is commenced or continued.

(126) The fluid presence sensor **948** located at the midpoint of the outlet chamber **1054** targets area **1168** and is used to control the amount of air that has accumulated in the outlet chamber **1054**. During normal operation, the fluid level in the outlet chamber **1054** should be maintained proximate the midpoint of the outlet chamber. If fluid is not detected by the fluid midpoint sensor **948**, and the pressure sensors **949** are reading a positive pressure, the control system opens the solenoid valve **951** to expel excess air that has accumulated in the outlet chamber **1054** until the fluid midpoint sensor **948** detects fluid (i.e., until the fluid level has increased to the midpoint of the outlet chamber **1054**). To avoid materially impacting the pressure monitoring and control function of the system **100**, the solenoid valve **951** may have a small orifice or restriction so that the excess air in the outlet chamber **1054** bleeds off at a low, controlled rate. Alternatively, the system **100** can average the fluid pressure readings so that the effects of any minor pressure decreases associated with the air expelling function are mitigated, or the system **100** can ignore the fluid pressure readings while the solenoid valve **951** remains open.

(127) The fluid presence sensor **950** located proximate the pressure port **1062** of the outlet chamber **1054** targets area **1169** to ensure proper operation of the pressure sensing function of the system **100**, which requires that a pocket of air be maintained between the fluid in the outlet chamber **1054** and the pressure sensors **949** of the sensing assembly **315**. The pressure of this pocket of air, which is monitored by the pressure sensors **949**, increases and decreases as a result of increases and decreases in the fluid pressure. If the fluid level reaches the hydrophobic filter **1065** that protects the pressure port **1062**, the control system may lose the ability to accurately monitor the fluid pressure. Accordingly, if the fluid pressure port sensor **950** senses fluid, the control system may disable pump **212**.

(128) Referring to FIGS. **6** and **14** through **18**, if the system **100** includes a main unit **102** with fluid warming capability, for example when configured for an operating room environment, the fluid conditioner **420** will generally be connected to the fluid warming cartridge component **422**. Joining the fluid conditioner **420** and fluid warming cartridge **422** together form cartridge assembly **419** and causes fluid connections to be made between the inlet chamber **1053** (FIG. **10**) of the fluid conditioner **420** and the first fluid path **629** (FIG. **6**) on a first side **1671** (FIGS. **16-18**) of the warming cartridge **422**. This connection also causes fluid connections between the second fluid path **631** (FIG. **6**) on the second side **1670** (FIGS. **16-18**) of the warming cartridge **422** and the outlet chamber **1054** (FIG. **10**) of the fluid conditioner **420**.

(129) The fluid warming cartridge **422** may include a rigid body **1472** (FIGS. **14-18**), a first thin flexible sheet **1473** (FIGS. **15-18**), and a second thin flexible sheet **1474** (FIGS. **15-18**). Referring to FIGS. **16-18**, the first flexible sheet **1474** is connected to a first side **1671** of the rigid body **1472** to define the first fluid flow path **629**, and the second flexible sheet **1473** is connected to a second side **1670** of the rigid body **1472** to define the second fluid flow path **631**. In the illustrated embodiment, the first and second fluid flow paths **629**, **631** are connected by a connector or tube **530** (FIGS. **14-15**). In other embodiments, the first and second flow paths may be connected by a channel that is integral to the warming cartridge **422**. The rigid body **1472** can be, for example, an injection molded body. The flexible side sheets **1473**, **1474** can be made of, for example, plastic

which is highly transmissive to IR to facilitate the fluid warming function. The rigid body **1472** and the flexible side sheets may be connected by gluing, laser welding, ultrasonic welding, or any other suitable means.

(130) The flexible side sheets **1473**, **1474** may be configured to expand and contract to effectively dampen fluid pulsations generated by the pump **212**, which allows the fluid delivered to the surgical site to be non-pulsatile. That is, although the system **100** may utilize a peristaltic pump which generates a pulsatile fluid flow, the fluid warming cartridge **422**, which is downstream of the peristaltic pump, may include thin, flexible side sheets **1473**, **1474** to at least partially define the fluid paths and expand and contract as the pressure of the fluid moving through the warming cartridge fluctuates to dampen the fluid pulsations. This damping of the fluid pulsations facilitates steady distention and good visualization during a surgical procedure.

(131) Referring to FIG. 6, in operation, fluid from the fluid supply bags or containers enters the fluid inlet chamber **1053** (FIG. 10) of the fluid conditioner **420** via port **625**, enters the fluid warming cartridge **422**, flows through the first elongated section of the fluid path **629** on a first side **1671** (FIGS. 16-18) of fluid warming cartridge **422**, exits the first elongated section of the fluid path and enters the second elongated section of the fluid path **631** on a second side **1670** of the fluid warming cartridge via connector **530**, exits the fluid warming cartridge **422** and enters fluid outlet chamber **1054** of the fluid conditioner **420**, and then exits the fluid outlet chamber **1054** for delivery to the surgical site via port **634**.

(132) The system **100** can control fluid temperature by monitoring the difference between the setpoint fluid temperature and the actual outlet fluid temperature sensed by temperature sensor **945** (FIG. 9) to adjust power to the IR lamp assemblies **737** (FIG. 7) in accordance with proportional integral control and scaling, which is based on the actual fluid flow rate and/or the difference between the actual fluid temperature sensed by the temperature sensor **944** aligned with the inlet chamber **1053** of the fluid conditioner **420** and the actual fluid temperature sensed by the temperature sensor **945** aligned with the outlet chamber **1054** of the fluid conditioner **420**.

Alternatively, other suitable open-loop and closed-loop control systems can be employed, such as, for example, proportional control, integral control, proportion-integral-derivative control, mathematical modelling, predictive function control, error squared control, and bang-bang control.

(133) In addition to the control scheme, the fluid warming efficiency can be enhanced by utilization of thin, flexible side sheets **1473**, **1474** (FIG. 15) of the fluid warming cartridge **422**, which may be highly transmissive to IR energy and an injection molded rigid body **1472** (e.g., a black injection molded body) that absorbs IR energy from the IR lamp assemblies **737** (FIG. 7) and radiates the IR energy back to the fluid. Additionally, the fluid warming efficiency can be enhanced by the elongated sections of the fluid warming cartridge **422** that define fluid paths **629**, **631**. The elongated sections of the fluid warming cartridge **422** can facilitate uniform heat distribution by introducing fluid into each section at or below the centerline and exiting fluid from each section at the top of the opposite end such that the fluid moves from a lower position to a higher position as it moves along each of the fluid paths **629**, **631**.

(134) Referring to FIG. 19, in certain embodiments, the fluid warming efficiency of system **100** can also be enhanced by pre-warming of the fluid containers **1901**. That is, air intake **1903** allows air to be drawn by fans **316** (FIG. 3) of heater assembly **314** (FIG. 3) into the main unit **102** to cool the heater assembly **314** and the main unit **102** during the fluid warming process, and this air becomes heated as a result of interacting with the heater assembly **314**. The heated air is then exhausted by the main unit **102** through exhaust openings **1905** and directed towards the fluid containers **1901** on each side of main unit **102** such that the fluid within the fluid containers is pre-warmed prior to being pumped through the fluid conditioner **420** (FIG. 6) and fluid warming cartridge **422** (FIG. 6).

(135) To guard against overtemperature conditions, the system **100** has low and high limits that disable the IR lamps **739** (FIG. 7) if the fluid temperature exceeds a low safety limit, and disables the IR lamps **739** and the pump **212** if the fluid temperature exceeds a high safety limit. In some

embodiments, independent of software, the system **100** employs a hardware circuit that includes the thermal cutoff sensor (“TCO”) **946** (FIG. **9**) to disable the IR lamps and the pump in an overtemperature condition exceeding the high limit. In some embodiments, the system **100** employs a cooling fan to remove heat to help prevent and/or mitigate overtemperature conditions. The cooling fan may be electronically controlled based on thermistor or other temperature sensor inputs and/or heating algorithm conditions that could lead to overtemperature (e.g. a rapid decrease in flow rate where full heating was required at maximum flow rates).

(136) Due in large part to the fluid warming function of the system **100**, which is intended to rapidly warm fluid up to the setpoint fluid temperature and to maintain the setpoint fluid temperature at high flow rates, the system **100**, for markets where the nominal supply voltage is 120V, must be connected to a dedicated 20-amp circuit. However, the system **100** can be configured for connection to a standard 15-amp circuit by utilizing lower wattage lamps and/or current limiting power to the lamps. In certain embodiments, fluid management system **100** is configured to operate on nominal supply voltage of 120 v or 240 v without requiring a change in lamps. For example, the system **100** can include a crossover circuit that includes a threshold detector **770** (FIG. **7A**) and a relay bank **771** (FIGS. **7B-7C**). Referring to FIG. **7**, in certain embodiments, the lamp assembly **737** includes four lamps **739** (e.g., two lamps on each side of the cartridge **419**). FIG. **7B** shows a circuit of a relay bank **771** for two lamps **739** of the lamp assembly **737** (e.g., two lamps **739** positioned on the same side of the cartridge **419**) with relay contacts **775**, **777** in a first position in which the lamps **739** are placed into a parallel configuration. FIG. **7A** shows a circuit for a threshold detector **770** that can cause the relay contacts **775**, **777** (FIGS. **7B-7C**) to move to a second position (as shown in FIG. **7C**) in which the lamps **739** are placed into a series configuration. While FIGS. **7B** and **7C** show a circuit for a relay bank **771** of two lamps **739** that are disposed on one side of the cartridge **719** shown in FIG. **7**, it should be understood that the circuit for the other two lamps **739** on the other side of the cartridge may be identical to the circuit shown in FIGS. **7B** and **7C**.

(137) Referring to FIG. **7A**, the threshold detector **770** has an AC line input **772** and an AC line input **774** that each connect to the input side of bi-directional photocoupler **776**, where the voltage being supplied to the heating assembly **314** is also applied to the inputs **772**, **774**. A first Zener diode **778** is positioned between the line input **772** and an input of the photocoupler **776**, and a second Zener diode **780** is positioned between the line input **774** and the input of the photocoupler **776**. The Zener diodes **778**, **780** prevent current flow from the inputs **772**, **774** through the photocoupler **776** until the peak voltage applied to the inputs **772**, **774** is greater than or equal to a predetermined amount. The photocoupler **776** includes a transistor output element **782** connected to voltage source **784**, where the transistor output element **782** is configured to operate between an “On” position and “Off” position with a line **786** that generates a control signal used to energize relay coils **788** that operate the relay bank **771**. The transistor output element **782** remains off when current is not moving through the input of the photocoupler **776**, and the transistor output element **782** turns on when the current flows through Zener diodes **778**, **780** and enters the input of the photocoupler **776**. When the transistor output element **782** turns on, current from the voltage source **784** energizes line **786** to generate the control signal to activate the coils **788a-d** and move corresponding contacts (e.g., contacts **775**, **777**) of the relay bank **771** from the first position (as shown in FIG. **7B**) to the second position (as shown in FIG. **7C**). In some embodiments, the control signal is applied to a Schmitt trigger **790** and transistor array **792** prior to energizing the coils **788a-d**. The Schmitt trigger **790** ensures that the voltage on line **786** is stable and exceeds a predetermined limit prior to being connected to the transistor array. The transistor array **792** energizes coils **788a-d** and moves the relay contacts **775**, **777** from the first position to the second position.

(138) Referring to FIG. **7B**, in the illustrated embodiment, the contacts **775**, **777** are in the first position when the lamps **739** are in a parallel configuration. When in the first position, the contacts

775, 777 are connected at points 4 and 5 such that both lamps 739 are in communication with the inlet 773 of the circuit. This allows half the current moving through inlet 773 to move along a first path 779 into one lamp 739 and half the current to move through the second path 781 (through the contact 775) and into the second lamp 739. In this configuration, the same voltage applied to the inlet 773 is individually applied across both of the lamps 739 such that each lamp 739 receives the voltage entering the circuit. For example, if 120 v is applied to inlet 773, the 120 v is connected to both the first path 779 and the second path 781 such that each lamp receives 120 v.

(139) Referring to FIG. 7C, when the threshold detector 770 (FIG. 7A) causes the contacts 775, 777 to move to the second position, an electrical connection is made between points 3 and 5 such that the lamps 739 are connected in a series configuration. In this configuration, all the current applied to the inlet 773 moves along a single path 783 (because the disconnection between points 4 and 5 prevents the current from moving directly through the contact 775 after entering the inlet 773) such that the current moves through one lamp 739, continues to move along the path 783 such that the current moves through the contact 775 at points 3 and 5, and the current then moves into the second lamp 739. Because both lamps 739 are disposed along a single path 783, the voltage applied to the lamps 739 is split between the number of lamps disposed on the path 783. As the illustrated embodiment includes two lamps 739, each lamp 739 receives half of the voltage entering the inlet 773. For example, if 240 v is applied to inlet 773, all the current flows along the single path 783 such that the voltage drop across one lamp 739 is 120 v and this dropped voltage is applied along the single path 783 such that the other lamp receives 120 v.

(140) The fluid management system 100 may be configured to provide accurate and reliable flow-based deficit monitoring for surgical procedures performed in an operating room environment. For example, FIGS. 20 through 49 illustrate an exemplary embodiment of a deficit module 104 and a single or multiuse deficit cartridge 2010 for the fluid management system shown in FIG. 1. The use of the deficit cartridge 2010 negates the need for canisters and avoids exposing the sensors and other durable components of fluid management system 100 to the fluid returning from the surgical site. The deficit module 104 works in combination with the control system of the main unit 102 and a single or multiuse tubing set that includes the deficit cartridge 2010 to measure and record the fluid volume being returned from the surgical site as it moves through deficit cartridge 2010. The fluid is pulled from the surgical site and both into and out of the deficit cartridge 2010 by a suction source (e.g., a vacuum pump integral or external to the system 100). In an alternative embodiment, fluid may be pulled from the surgical site and pushed into the deficit cartridge 2010 by a more positive pressure than present inside deficit cartridge 2010 (e.g., a peristaltic pump integral or external to the system 100 inserted in-line between the surgical site and the deficit cartridge 2010 and configured to create suction at the surgical site and positive pressure at the inlet to the deficit cartridge 2010) and pulled from the deficit cartridge 2010 by a more negative pressure than present inside deficit cartridge 2010 (e.g., a vacuum pump integral to or external to the system 100 or a sufficiently positive pressure inside the deficit cartridge 2010 due to the positive pressure created by the peristaltic pump to push the fluid out of the deficit cartridge 2010 to ambient pressure).

(141) Referring to FIGS. 20 through 23, the deficit cartridge 2010 is inserted into the deficit module 104. The deficit cartridge 2010 may include a front section 2218 that aligns with an opening 2220 (FIGS. 22-23) of the deficit module 104 and a raised portion 2222 (FIGS. 22-23) that allows for a user to easily grasp the deficit cartridge 2010 to remove the cartridge from the deficit module 104. The deficit cartridge 2010 includes one or more inlet openings 2012, 2014 that are configured to connect to one or more fluid return tubes of the tubing set such that fluid can move from the surgical site and into the deficit cartridge 2010. The deficit cartridge also includes at least one vacuum opening 2016 that is configured to connect to an evacuation tube such that fluid moves through the evacuation tube after moving through the deficit cartridge 2010. The evacuation tube is connected to the suction source such that a vacuum pressure is supplied to the deficit cartridge to pull fluid from the surgical site into and out of the deficit cartridge 2010. The fluid return and

evacuation tubes may be manually connected to the deficit cartridge **2010** after insertion of the deficit cartridge into the deficit module **104**.

(142) The control system of the main unit **102** is configured to determine a deficit of fluid provided to the surgical site and returned from the surgical site by comparing the volume of fluid moving through the deficit cartridge **2010** to the volume of fluid being supplied to the surgical site. The control system may calculate the volume of fluid being supplied to the surgical site by, for example, monitoring the weight of the fluid supply bags or containers (e.g., by using the hanging members **116** that are operatively connected to load cells), and/or counting the rotations of a peristaltic pump.

(143) Referring to FIGS. **24** through **28**, insertion of the deficit cartridge **2010** into the deficit module **104** causes a manifold connection assembly **2424** (FIGS. **27-28**) to engage the deficit cartridge **2010** and connect, via a pump manifold assembly (e.g., pump manifold assembly **3513** shown in FIG. **37**), positive and negative pressure pumps (**3515** and **3517**, respectively, of FIGS. **35-36**) and a pressure sensor (not shown) of the deficit module **104** to pneumatically operated diaphragm regulators/valves (e.g., regulators/valves **2628**, **2630**, **2632**, **2634** shown in FIG. **26**) and a pressure sensing area **2636** (FIG. **26**) of the deficit cartridge **2010**. The manifold connection assembly **2424** has a plurality of connectors (e.g., connectors **4510**, **4511**, **4512**, **4513** shown in FIGS. **45** and **49**) for receiving corresponding ports **2540** for each of the regulators/valves and the pressure sensing area of the deficit cartridge **2010**. Referring to FIGS. **27-28** (and FIGS. **42-49**), the connectors of the manifold connection assembly **2424** can be configured to be moved between an engaged or connected state and a disengaged or disconnected state relative to the ports **2540** by a mechanical or electromechanical mechanism **2726** (e.g., a manual lever, a Pancake cylinder, or other type of pneumatic, mechanical, or electromechanical actuator). The ports **2540** of the deficit cartridge **2010** may include an O-ring that allows for a hermetically sealed connection between the manifold connection assembly **2424** and the deficit cartridge **2010**. Insertion of the deficit cartridge **2010** into the deficit module **104** also causes one or more non-contact fluid sensors **2742** (FIGS. **30-31**) of the deficit module **104** to align with desired locations of the deficit cartridge **2010**.

(144) The connectors (e.g., connectors **4510**, **4511**, **4512**, **4513** shown in FIGS. **45** and **49**) of the manifold connection assembly **2424** may be configured to account for any manufacturing or assembly tolerances in the ports **2540** of the deficit cartridge **2010**. That is, the connectors may be configured to move to ensure alignment with the corresponding ports **2540** of the deficit cartridge **2010** to account for minor differences in the location of the ports **2540** resulting from manufacturing and assembly of the deficit cartridge **2010**. For example, referring to FIGS. **28A-28C**, in certain embodiments, a connector **4512** (also shown in FIGS. **45** and **49**) of the manifold assembly **2424** may be a separate component that is connected to the manifold assembly **2424** by an attachment element **2815** (e.g., an E-clip), and a receiving assembly **3511** (also shown in FIGS. **38-39**) of the system **100** may include an opening **3828** (also shown in FIGS. **38-39**) that is larger than the diameter of the connector **4512** for receiving the connector **4512** such that the connector **4512** can move within the opening **3828**.

(145) Referring to FIG. **28A**, the manifold connection assembly **2424** is shown in a disengaged position with the port **2540** of the deficit cartridge **2010**. Activation of the mechanism **2726** (FIG. **28**) causes the manifold connection assembly **2424** to move in the direction **M** such that the connector **4512** engages the port **2540** of the deficit cartridge **2010**. FIG. **28B** shows the initial engagement between the connector **4512** and the port **2540**, and FIG. **28C** shows the completed engagement between the connector **4512** and the port **2540**. Referring to FIG. **28B**, the port **2540** of the deficit cartridge **2010** is not centered with the connector **4512**, which causes the port **2540** to engage an edge of an inlet **2813** of the connector **4512**. The large opening **3828** of the receiving assembly **3511** allows for the connector **4512** to move within the opening **3828** and align with the port **2540**. That is, referring to FIG. **28C**, continued movement of the manifold connection assembly **2424** in the direction **M** causes the port **2540** to align with and move into the connector

4512. In certain embodiments, the inlet **2813** of the connector **4512** is tapered to facilitate movement of the port **2540** into the connector **4512**. The connection described above between the connector **4512** and the port **2540** allows for an easy and automatic connection between the deficit cartridge **2010** and the system **100** (e.g., via the deficit module **104**). While FIGS. **28A-28C** only show the connection between connector **4512** and a port **2540** of the deficit cartridge **2010**, it should be understood that the other connectors (e.g., connectors **4510**, **4511**, **4512**, **4513** shown in FIGS. **45** and **49**) may be configured to connect to the ports **2540** of the deficit cartridge **2010** in the same manner described in FIGS. **28A-28C**.

(146) Referring to FIGS. **29-34**, the deficit cartridge **2010** may include a single chamber **2944** with three sections **2946**, **2948**, **2950** that are fluidically connected. The three sections include a fill section **2946**, a measure section **2948**, and an evacuation section **2950** that are fluidly connected at all times as the system **100** alternates between “Fill/Measure” and “Fill/Evacuation” Cycles, which allows the pressure gradient across the three sections to be minimized or substantially equal.

(147) The fill section **2946** is fluidly connected to the inlet openings **2012**, **2014** such that fluid returning from a surgical site can move into the fill section **2946** through the inlet openings **2012**, **2014**. The evacuation section **2950** is fluidly connected to the vacuum port **2016** such that a suction source can supply a vacuum pressure to the deficit cartridge **2010** that causes fluid to move from the surgical site, into the deficit cartridge **2010** through the inlet openings **2012**, **2014**, and exit the deficit cartridge **2010** through the vacuum port **2016**. In other embodiments, a pump in-line between the surgical site and port **2012** and/or port **2014** (e.g., a peristaltic pump) may pull fluid from the surgical site and push the fluid through the inlet openings **2012** and/or **2014** and out of the deficit cartridge **2010** through the vacuum port **2016**. Alternatively, a pump in-line between the surgical site and port **2012** and/or port **2014** (e.g., a peristaltic pump) may pull fluid from the surgical site and push the fluid through the inlet openings **2012** and/or **2014** while a separate suction source supplies vacuum pressure to pull fluid out of the deficit cartridge **2010** through the vacuum port **2016**.

(148) One or more inlet valves **2628**, **2630** may be positioned at the inlet openings **2012**, **2014** and configured to close to prevent fluid from entering chamber **2944** to avoid overfill conditions. In certain embodiments, the valves **2628**, **2630** are pneumatically-operated diaphragm valves that are connected to a pump assembly (e.g., an assembly including positive pressure pump **3515** and negative pressure pump **3517** shown in FIGS. **35-36**) of the deficit module **104** such that the pump assembly can move the valves **2628**, **2630** between the open and closed positions. In certain embodiments, a negative pressure pump of the pump assembly opens the diaphragm valves and a positive pressure pump of the pump assembly assists closing of the diaphragm valves with positive back pressure.

(149) The pneumatically-operated diaphragm valves **2628**, **2630** may also work in combination with the pump assembly (e.g., an assembly including positive pressure pump **3515** and negative pressure pump **3517** shown in FIGS. **35-36**) to act as a pressure regulator that regulates the vacuum pressure being supplied to the surgical site. That is, the control system of the fluid management system **100** may be configured to adjust the amount of pressure applied to the valves **2628**, **2630** by the pump assembly, which adjusts the threshold pressure required to displace flexible membrane **2956** and allow fluid to flow through **2628**, **2630**, and thereby allows the control system to control the amount of vacuum pressure supplied to the surgical site via the suction source that is connected to the vacuum port **2016**. For example, referring to FIG. **29**, the valves **2628**, **2630** may each include a housing component **2958** that defines a chamber **2959**, where the chamber **2959** is connected to the pump assembly of the deficit cartridge. A flexible membrane **2956** is disposed in the chamber **2959** and movable within the chamber **2959** by the pressure pumps. When the valves **2628**, **2630** are in the closed position, the membrane **2956** engages the chamber **2944** of the deficit cartridge **2010** to fluidly isolate the fill section **2946** from the inlet openings **2012**, **2014**. The pressure pumps are configured to move the flexible membrane **2956** within the chamber to open

the valves **2628**, **2630**, and the pump assembly can adjust the size of the opening by creating a desired pressure differential between the pressure supplied by the pump assembly and the vacuum level within the chamber **2944** of the deficit cartridge **2010**. The membrane **2956** can be made of, for example, neoprene, silicone, natural rubber, nitrile, EPDM, or any other suitable material. The valves **2628**, **2630** may also have a hydrophobic filter **2960** to prevent fluid from traveling to the pump assembly in case of a tear or other failure of the flexible membrane. In other words, the valves **2628**, **2630** work similar to the pressure regulator described with respect to FIGS. 52-73 of the present application to regulate the vacuum pressure supplied to the surgical site.

(150) In the illustrated embodiment, the fill section **2946** is positioned at a top portion of the chamber **2944**, and the measure section **2948** is positioned below the fill section **2946**. A valve **2632** is positioned in an opening between the fill and measure sections **2946**, **2948** and is movable between an open position and a closed position. When the valve **2632** is in the open position, the fill section **2946** and the measure section **2948** are fluidly connected such that fluid in the fill section **2946** can move into the measure section **2948** via gravity. One or more sensors of the deficit module **104** are used to measure the fluid within the measure section **2948**. In certain embodiments, the measure section includes a main area **3276** (FIG. 32) and a narrow area **3277** (FIG. 32) positioned above the main area **3276**, where the volume of fluid capable of being disposed in these areas **3276**, **3777** are known by the system **100** such that the system can determine the volume of fluid moving through the measure section **2948**. The measuring of fluid within the measure section **2946** will be described in more detail below. The evacuation section **2950** is positioned below the measure section **2948**, and a valve **2634** is positioned in an opening between the measure and evacuation sections and is movable between an open position and a closed position. When the valve **2634** is in the open position, the measure section **2948** and the evacuation section **2950** are fluidly connected such that fluid in the measure section **2948** can move into the evacuation section **2950** via gravity. In the illustrated embodiment, the valves **2632**, **2634** are pneumatically-operated diaphragm valves that are connected to a pump assembly (e.g., an assembly including pumps **3515**, **3517** shown in FIGS. 35-36) of the deficit module **104** such that the pump assembly moves the valves **2632**, **2634** between the open and closed positions.

(151) Referring to FIG. 32, the valves **2628**, **2630**, **2632**, **2634** may include a flexible membrane (e.g., flexible membrane **2956** shown in FIG. 29) that is movable between an engaged position and a disengaged position with openings **3290** of the deficit cartridge **2010**. That is, a portion of the openings **3290** fluidly connect the inlets **2012**, **2014** to the fill section **2946**, another portion of the openings **3290** fluidly connect the fill section **2946** to the measure section **2948**, and another portion of the openings **3290** fluidly connect the measure section **2948** to the evacuation section **2950**. The flexible membranes of the valves **2628**, **2630**, **2632**, **2634** engage the openings **3290** to prevent movement of fluid between the inlets/sections, and disengage at least a portion of the openings **3290** to allow movement of flow between the inlets/sections. The size and spacing of the openings **3290** may be configured to prevent extrusion of the flexible membranes through the openings **3290** when positive pressure is applied to the valves **2628**, **2630**, **2632**, **2634**. The size and spacing of the openings **3290** may vary based on the elasticity and/or thickness of the material of the flexible membrane. The number of openings **3290** associated with each valve **2628**, **2630**, **2632**, **2634** are configured to ensure adequate flow of fluid through the deficit cartridge **2010**. In certain embodiments, the combined surface area of the openings on each side of the valves **2628**, **2630**, **2632**, **2634** is substantially equal to an inner cross-sectional area of tubing that is attached to the inlet ports **2012**, **2014** of the deficit cartridge **2010**. Because gravity is the dominant force acting on the fluid to cause the fluid to move between the sections **2946**, **2948**, **2950** of the chamber **2944**, in some embodiments, the number of openings **3290** corresponding to the valves **2632**, **2634** is configured to be large enough to allow sufficient flow through the valves **2632**, **2634** to achieve high flow rates. For example, the number of openings **3290** corresponding to each valve **2632**, **2634** may be configured to achieve a target flow rate of greater than or equal to 1200 ml/min

through the chamber **2944** without stopping flow from the surgical site. In certain embodiments, because the measure section **2948** of the chamber **2944** may be both filled and emptied while fluid continuously flows from the surgical site, the fluid flow rate through the valves **2632**, **2634** may be at least twice the target flow rate through the chamber **2944**.

(152) In the illustrated embodiment, the chamber **2944** includes a channel **2952** that fluidly connects the fill section **2946** to the evacuation section **2950** and a narrow portion **3277** that fluidly connects the fill section **2946** to the measure section **2948**. The channel **2952** and narrow portion **3277** allow the fill, measure, and evacuation sections **2946**, **2948**, and **2950** to be fluidly connected at all times, including when one or both of the valves **2632**, **2634** are in the closed position. This fluid connection between the fill, measure, and evacuation sections **2946**, **2948**, and **2950** via the channel **2952** and narrow portion **3277** allows the pressure gradient across the three sections of the chamber **2944** to be minimized or substantially equal such that the fluid is not caused to move within the chamber **2944** by a pressure source, but rather the fluid can move within the chamber **2944** due to gravity. In certain embodiments, the deficit cartridge includes a wall **3252** that is positioned to prevent fluid from entering the channel **2952** and bypassing the measure section **2948**. In an alternative embodiment, rather than the chamber **2944** including the channel **2952**, the deficit cartridge **2010** can include a connector or tube (e.g., similar to tube **841** shown in FIG. **8** for the fluid conditioner **420**) that fluidly connects the fill section **2946** to the evacuation section **2950** such that the fill, measure, and evacuation sections **2946**, **2948**, and **2950** are fluidly connected at all times. While the illustrated embodiment shows the three sections **2946**, **2948**, **2950** being in a stacked configuration, in an alternative embodiment, these three sections can be in a side-by-side configuration, as long as the fluid can travel from the fill section **2946** to the measure section **2948** to the evacuation section **2950** via gravity.

(153) In various embodiments, the deficit cartridge **2010** includes a waste vacuum level sensing and regulation port **2636** for connecting to a solenoid valve which is open to ambient on one side and a pressure sensor of the deficit module **104**. The control system of the fluid management system **100** is capable of sensing the vacuum level of the chamber **2944** via the pressure sensor of the deficit module **104** and opening the solenoid valve to atmospheric pressure to down regulate the vacuum pressure being supplied to the deficit cartridge **2010** via the suction source. Referring to FIG. **29**, the port **2636** may include a housing component **2973** that defines a chamber **2975**, where the chamber **2975** is connected to the solenoid valve and pressure sensor of the deficit module **104**. A flexible membrane **2977** is disposed in the chamber **2975** and has an opening **2979** that enables pressure measurement of the pressure in chamber **2944** and exposes chamber **2944** to ambient when the solenoid valve is open. The membrane **2977** can be made of, for example, neoprene, silicone, natural rubber, nitrile, EPDM, or any other suitable material. The port **2971** may also have a hydrophobic filter **2981**.

(154) Referring to FIGS. **30** through **32**, the deficit cartridge **2010** is aligned with one or more sensors **2742** (e.g., sensors **3062**, **3064**, **3066**, **3068**, **3070** shown in FIGS. **30-31**) of the deficit module **104** such that the control system of the fluid management system **100** can use the sensors **2742** to detect a volume of fluid moving through the chamber **2944** of the deficit cartridge **2010** and/or detect any potential problems with fluid flow through the deficit cartridge (e.g., potential overflow of fluid within section **2946** of the chamber **2944**). In the illustrated embodiment, the one or more sensors **2742** include a first fluid presence sensor **3062**, a second fluid presence sensor **3064**, a third fluid presence sensor **3066**, a fourth fluid presence sensor **3068**, and a fifth fluid presence sensor **3070**. Referring to FIGS. **32** through **34**, in the illustrated embodiment, the first and second fluid presence sensors **3062**, **3064** are aligned with first and second areas **3271**, **3272**, respectively, within the fill section **2946**. These fluid presence sensors **3062**, **3064** are used by the control system to close one or both of the inlet valves **2628**, **2630** if the fluid within the fill section **2946** reaches the first and second areas **3271**, **3272**. The third fluid presence sensor **3066** is aligned with a third area **3273** in the measure section **2948** of the chamber **2944** and is used by the control

system to switch from Fill/Measure cycle to the Fill/Evacuation cycle and determine a volume of fluid within the measure section **2948** prior to switch to the Fill/Evacuation cycle. The fourth fluid presence sensor **3068** is aligned with a fourth area **3274** within the measure section **2948** and is used by the system to switch from the Fill/Evacuation cycle to the Fill/Measure cycle. The fifth fluid presence sensor **3070** is aligned with a fifth area **3275** within the measure section **2948** and is used by the system to provide more accurate real-time fluid volume measurement in measure section **2948** and to determine a volume of fluid in the measure section **2948** after the procedure is completed or the type of fluid being monitored for recording a fluid deficit is changed to provide a more accurate fluid deficit calculation for the fluid. While the illustrated embodiment shows the deficit module **104** having five fluid presence sensors for detecting fluid flow conditions and volume within the chamber **2944** of the deficit cartridge, it should be understood that any other suitable number of fluid presence sensors can be used by the deficit module to detect fluid flow conditions and volume. The target areas **3271-3275** may include walls partially surrounding them to mitigate the effects of fluid turbulence on the accuracy of the sensor readings and any flexing of the film **2980**.

(155) Referring to FIG. **29**, in the illustrated embodiment, the deficit cartridge **2010** includes a rigid body **2978** and a film **2980**. The rigid body **2978** partially defines the various sections **2946**, **2948**, **2950** and the channel **2952** and narrow portion **3277** of the chamber **2944**, and the film **2980** is attached to the rigid body **2978** to enclose the chamber **2944**. The rigid body **2978** can be, for example, an injection molded body or any other suitably rigid body. The film **2980** is configured to allow the one or more sensors of the deficit module **104** to detect characteristics of the fluid through the film without contacting the fluid. The film **2980** can be, for example, a plastic film. The film **2980** can be attached to the rigid body **2978** with mechanical fasteners or by gluing, laser welding, vibration welding, ultrasonic welding, or any other suitable means. In alternative embodiments, the deficit cartridge **2010** does not include film **2980**, but rather is made of an injection molded vessel that is capable of having the one or more sensors of the deficit module **104** detect characteristics of the fluid through the vessel without contacting the fluid. In other alternative embodiments, vessel may be cast or machined out of a material that is capable of being cleaned and reused.

(156) FIGS. **33** and **34** illustrate the Fill/Measure cycle and the Fill/Evacuation cycle for the deficit cartridge **2010**. Referring to FIG. **33**, during the Fill/Measure cycle, fluid returning from the surgical site is pulled from the surgical site into the fill section **2946** of the deficit cartridge **2010** through inlet ports **2012**, **2014** via a vacuum pressure from a suction source that is attached to vacuum port **2016**. The diaphragm operated valve **2632** is in the open position, which allows fluid to travel from the fill section **2946** to the measure section **2948** via gravity. The diaphragm operated valve **2634** is in the closed position, which prevents fluid from the measure section **2948** from moving into the evacuation section **2950**. The Fill/Measure cycle continues until the fluid level in the measure section **2948** has reached a predetermined level as sensed by the fluid presence sensor **3066** (FIGS. **30-31**) of the deficit module **104** that targets area **3273**. In the illustrated embodiment, the targeted area **3273** is disposed in a narrow portion **3277** (FIG. **32**) of the measure section **2948** that extends above from the main portion **3276** (FIG. **32**) of the measure section **2948**. The volume of fluid in the narrow portion **3277** is small compared to the volume of fluid in the main portion **3276** of the measure section **2948** and, therefore, variables including fluid flow rates and turbulence (which can affect the accuracy of the sensed fluid level) do not materially affect the overall accuracy of the measuring function. In certain embodiments, a ratio of the volume of the main portion **3276** to a volume of the narrow portion can be greater than or equal to 5 to 1, such as greater than or equal to 20 to 1, such as greater than or equal to 50 to 1, such as greater than or equal to 75 to 1, such as greater than or equal to 90 to 1, such as greater than or equal to 100 to 1. In an exemplary embodiment, the ratio of the volume of the main portion **3276** to the volume of the narrow portion can be about 100 to 1. The volume of fluid within the main portion **3276** and

narrow portion **3277** of the measure section **2948** are known by the system **100**, which allows the system to record the volume of fluid within the measure section for each time the Fill/Measure cycle occurs. The system **100** records the volume and then transitions to the Fill/Evacuation cycle. (157) Referring to FIG. **34**, during the Fill/Evacuation cycle, the diaphragm operated valve **2632** is moved to the closed position, which prevents fluid from the fill section **2946** from moving into the measure section **2948**. The diaphragm operated valve **2634** is moved to the open position, which allows the fluid that was measured in the measure section **2948** during the Fill/Measure cycle to move into the evacuation section **2950** via gravity. The fluid entering the evacuation section **2950** is then evacuated through the vacuum port **2016**, via the attached suction source, and the fluid is moved to the facility's waste disposal system via indirect-to-drain or direct-to-drain methods. To evacuate fluid from the evacuation section, the evacuation cycle relies upon a vacuum pressure differential between the vacuum pressure provided by the suction source and the down-regulated vacuum pressure inside of the chamber **2944** (as regulated via the pressure regulation and sensing port **2636**). When the fluid presence sensor **3068** (FIGS. **30-31**) that targets area **3274** detects no remaining fluid in the measure section **2948**, the system **100** transitions back to the Fill/Measure cycle. The alternation between the Fill/Measure cycle and the Fill/Evacuation cycle continues until the procedure is completed, and the control system determines the fluid deficit of the fluid based at least partially on the various volume measurement recordings taken during the various Fill/Measure cycles.

(158) The movement of fluid from the fill section **2946** to the measure section **2948** and the evacuation section **2950** is accomplished with gravity, as opposed to external suction or pressure sources. In these embodiments, the valves may be sized to minimize resistance and thereby facilitate high flow rates with relatively low forces. The pneumatically-actuated diaphragm valves **2628**, **2630** accomplish the allowance or stoppage of flow into the deficit cartridge **2010**, and the pneumatically-actuated diaphragm valves **2632**, **2634** accomplish the allowance and stoppage of flow between the sections, **2946**, **2948**, **2950**, by setting the pneumatic control pressure by the pressure pumps **3515**, **3517** (FIGS. **35-36**) of the deficit module **104** to a more positive gauge pressure than the combination of 1) the highest pressure expected on either wetted side of the valve, and 2) any additional pressure required to account for the additional force from the spring coefficient of the valve membrane.

(159) To guard against overflow conditions, the deficit module **104** may have a fluid presence sensor **3064** (FIGS. **30-31**) that targets area **3272**, and the control system may be configured to close the fluid return valve **2628** (e.g., the valve connected to the underbody drape and/or floor suction at the surgical site) if the fluid presence sensor **3064** detects fluid at the target area **3272**. This ensures that the fill section **2946** does not overflow and flow into the measure section **2948** through narrow portion **3277** or the evacuation section **2950** through the channel **2952**, and ensures that the remaining capacity of the fill section remains available to receive fluid returning from the surgical instrument at the surgical site so as not to interrupt the surgical procedure. The deficit module may also have a fluid presence sensor **3062** (FIGS. **30-31**) that targets area **3271**, and the control system may be configured to close the fluid return valve **2630** (e.g., the valve connected to a surgical instrument at the surgical site) if the presence sensor **3062** detects fluid at the target area **3271**. In an alternative embodiment, the valve **2628** can be connected to the surgical instrument at the surgical site, and the valve **2630** can be connected to the underbody drape and/or floor suction at the surgical site.

(160) To provide end of procedure fluid deficit accuracy (assuming the end of the surgical procedure does not coincide with the end of a Fill/Measure or Fill/Evacuation cycle), the deficit module **104** may include one or more midpoint fluid presence sensors (e.g., sensor **3070**) that target one or more areas (e.g., area **3275**) to provide more accurate real-time measurement of the fluid in measure section **2948** and to measure the fluid in the measure section **2948** at the end of a surgical procedure or after the type of fluid being used during the surgical procedure has been changed.

(161) FIGS. 35 through 48 illustrate an exemplary embodiment of a deficit module 104 that can be used with the fluid management system 100 shown in FIG. 1 and the deficit cartridge 2010 shown in FIGS. 20-34. Referring to FIG. 35, the deficit module 104 may include a deficit cartridge receiving assembly 3511 for receiving the deficit cartridge 2010, one or more sensors 2742 for sensing characteristics of fluid moving through the deficit cartridge without contacting the fluid, a pump assembly 3514, a pump manifold assembly 3513, a manifold connection assembly 2424 for connecting the deficit cartridge 2010 to the pump assembly 3514 and a solenoid and pressure sensor (via the pump manifold assembly 3513), a pneumatic mechanism 2726 that moves the manifold connection assembly between an engaged position (e.g., as shown in FIGS. 47-49) and a disengaged position (e.g., as shown in FIGS. 43-45) with the deficit cartridge 2010, and a printed circuit board (PCB) 3519.

(162) Referring to FIGS. 38 and 39, the deficit cartridge receiving assembly 3511 includes a base 3821 having a slot or opening 3820 for receiving the deficit cartridge 2010 (FIGS. 30-34). The receiving assembly 3511 also includes one or more walls or components 3822-3826 that substantially separate the deficit cartridge 2010 from the remainder of the components within the interior of the deficit module 104 when the deficit cartridge 2010 is disposed within the receiving assembly 3511. The walls or components 3822-3826 and base 3821 can be connected by one or more fasteners 3827 to create the receiving assembly 3511. A first wall 3822 of the receiving assembly 3511 can be configured to hold the one or more sensors 2742 for sensing characteristics of fluid moving through the deficit cartridge 2010. In the illustrated embodiments, the one or more sensors 2742 are capacitive sensors that detect fluid presence. However, other fluid level or presence sensing technologies could be utilized including infrared sensors, laser sensors, optical sensors, electro-mechanical sensors (e.g. float with mechanical toggle switch actuation, piezo-electric pressure sensors, etc.), inductive sensors, ultrasonic sensors, or any other suitable sensors.

(163) A second wall 3823 of the receiving assembly 3511 can include a plurality of openings 3828 for receiving connectors (e.g., connectors 4510-4513 shown in FIGS. 45 and 49) of the manifold connection assembly 2424 such that the pump assembly 3514 can be operatively connected to the diaphragm valves and pressure port of the deficit cartridge 2010, as discussed in more detail below with references to FIGS. 42-49. Referring to FIG. 39, the manifold connection assembly 2424 can be connected to or positioned adjacent to the wall 3823 of the receiving assembly 3511, and the pneumatic mechanism 2726 can be connected to the manifold assembly 2424 by one or more fasteners and to the deficit module 104 by a connection element or plate 3930.

(164) Referring to FIGS. 35 and 36, in the illustrated embodiment, the pump assembly 3514 includes a positive pressure pump 3515 and a negative pressure pump 3517, where the pump assembly 3514 is connected to the pneumatic mechanism 2726 and the connectors of the manifold connection assembly 2424 via the pump manifold assembly 3513. The positive pressure pump 3515 provides pressure to the pneumatic cylinder 2726 to move the manifold connection assembly 2424 between the engaged position (e.g., as shown in FIGS. 47-49) and the disengaged position (e.g., as shown in FIGS. 43-45) with the deficit cartridge 2010. The positive pressure pump 3515 also expedites closing or augments closing force of the diaphragm valves of the deficit cartridge 2010. The negative pressure pump 3517 provides a vacuum pressure to the diaphragm valves of the deficit cartridge 2010 to move the diaphragm valves to the open position.

(165) Referring to FIG. 37, in the illustrated embodiment, the pump manifold assembly 3513 includes a plurality of accumulators 3732 and solenoid valves 3734 for regulating pressure provided by the pumps 3515, 3517 and the opening and closing of the diaphragm valves of the deficit cartridge 2010. In certain embodiments, the accumulators 3732 vary the positive and negative pressures to: (1) allow for the use of smaller pressure pumps that can deliver the necessary flow rates; and (2) aid pressure regulation by reducing the impact of introducing air via a “bang-bang” control scheme (i.e., an open valve, close valve control scheme). In the illustrated embodiment, the pump manifold assembly 3513 includes five accumulators 3732 (e.g., holes that

extend through the assembly **3515**) that are capped off at the top and bottom by caps **3735**). The solenoid valves **3734** of the pump manifold assembly **3513** may connect to the connectors of the manifold connection assembly **2424** via tubing.

(166) Referring to FIGS. **42-49**, the pneumatic mechanism **2726** is shown moving the manifold connection assembly **2424** between a disengaged position (FIGS. **43-45**) with the deficit cartridge **2010** and an engaged position (FIGS. **47-49**) with the deficit cartridge **2010**. Referring to FIGS. **43-45**, when in the disengaged position, the ports **2540** for the diaphragm valves and pressure port of the deficit cartridge **2010** are not engaged by the connectors **4510-4513**. Referring to FIGS. **47-49**, the manifold connection assembly **2424** is moved to the engaged position in the direction D (FIG. **49**) by the pneumatic mechanism **2726** such that the connectors **4510-4513** engage a corresponding port **2540** of the deficit cartridge **2010**. When the connectors **4510-4513** are engaging the ports **2540** of the deficit cartridge **2010**, the pump assembly **3514** is operatively connected to the deficit cartridge **2010** such that the pump assembly **3514** can move the diaphragm valves of the deficit cartridge between the open and closed positions and the vacuum pressure supplied to the deficit cartridge **2010** can be sensed by a pressure sensor in the deficit module and down regulated by opening a solenoid to atmosphere.

(167) Referring to FIGS. **35-49**, in certain embodiments, insertion of the deficit cartridge **2010** into the receiving assembly **3511** of the deficit module **104** causes all of the internal connections between the deficit cartridge **2010** and the various components of the deficit module **104**. For example, insertion of the deficit cartridge **2010** into the receiving assembly **3511** causes the pneumatic mechanism **2726** to move the manifold connection assembly to the engaged position with the deficit cartridge (e.g., as shown in FIGS. **47-49**) and operatively connect the pump assembly **3514** to the deficit cartridge. Insertion of the deficit cartridge **2010** into the receiving assembly **3511** also causes the one or more non-contact sensors **2742** to be aligned with the chamber **2944** (FIG. **29**) of the deficit cartridge **2010**. These automatic connections between the deficit cartridge **2010** and the deficit module are advantageous because it limits the amount of connections a user has to make with respect to the deficit cartridge **2010**. That is, after inserting the deficit cartridge **2010** into the deficit module **104**, a user only needs to connect fluid return line(s) to inlet openings **2012**, **2014** (FIG. **20**) of the deficit cartridge **2010** and an evacuation line to a vacuum opening **2016** (FIG. **20**) of the deficit cartridge.

(168) Referring to FIG. **50**, in certain embodiments, the system **100** may be used for gynecological, urological, and orthopedic procedures that are performed in operating rooms that are equipped with third party suction and fluid collection devices. In these embodiments, the system **100** can be configured to include a main unit **102** (e.g., any main unit **102** described in the present application) and a deficit module **104** (any deficit module **104** described in the present application), but not include the fluid suction and collection module **106**. In this embodiment, a tubing set that includes deficit cartridge **2010** (FIGS. **20-34**) can be used in combination with the main unit **102** and the deficit module **104** to determine a fluid deficit of a fluid during a surgical procedure.

(169) Referring to FIG. **51**, in some situations, gynecological, urological, and orthopedic procedures are performed in operating rooms in which the facility prefers or requires “direct-to-drain” disposal of fluids returning from the surgical site. In these situations, the facility often prefers or requires the volume of the fluid being returned from the surgical site be recorded. In some embodiments, the system **100** can be configured to include a fluid flow monitoring and evacuation module **5101** that includes features of the deficit module **104**. The fluid flow monitoring and evacuation module **5101** can work in combination with the central suction system of the facility, main unit **102**, and a tubing set that includes a deficit cartridge **2010** (FIGS. **20-34**) or other similar cartridge to determine a fluid volume returning from the surgical site and entering a waste disposal system of the facility, as well as a fluid deficit for the surgical procedure. The fluid flow monitoring and evacuation module **5101** may communicate with the main unit **102** via Bluetooth or other wired or wireless means to measure, record, and display the return fluid volume and/or fluid

deficit for the surgical procedure.

(170) While the fluid flow monitoring and evacuation module **5101** is described as working in combination with the main unit **102** of the fluid management system **100**, it should be understood that the fluid flow monitoring and evacuation module **5101** may also function as a stand-alone fluid flow monitoring and evacuation module capable of communicating with other equipment of the facility via Bluetooth or other wired or wireless means. After recording the fluid volume, module **5101** can then dispose of the fluid directly into a waste disposal system of the facility. In certain embodiments, the fluid flow monitoring and evacuation module **5101** can be a wall mounted unit or a cart mounted unit. In some embodiments, the fluid flow monitoring and evacuation module **5101** can include an integrated suction source to work in combination with, or in place of, the central suction system of the facility. As use of the deficit cartridge **2010** or other similar cartridge isolates the fluid returning from the surgical site from the components (e.g., sensors, pumps, etc.) of the fluid flow monitoring and evacuation module **5101**, circulation of cleaning solution through the fluid flow monitoring and evacuation module after each procedure is not necessary, which enhances procedure efficiency.

(171) The flow-based deficit monitoring feature of the system **100** (e.g., the combination of the deficit module **104** and the deficit cartridge **2010**, or the fluid flow monitoring and evacuation module **5101**) enables accurate and reliable fluid deficit monitoring that is cost-effective due to the single-use nature of the deficit cartridge **2010** and the elimination of canisters. This feature also enhances procedure efficiency as interruptions associated with setting up, connecting, changing, and discarding of the canisters will also be eliminated, as well as cleaning the deficit module and/or monitoring and evacuation module after each procedure. In alternative embodiments, the deficit cartridge **2010** may be configured for multi-procedure use.

(172) In certain situations, the fluid management system **100** may be connected to an external pressure source (e.g., a suction source) that is used to pull fluid from the surgical site. As external suction sources are usually set to high vacuum levels in an operating room environment, down-regulation of the vacuum pressure provided by the external suction source may be necessary for proper operation of certain fluid outflow regulation, deficit monitoring, and/or collection functions. Down-regulation of a vacuum pressure provided by an external suction source can be accomplished via a manually or electronically-controlled regulator (“Regulator”) provided it is isolated from the biohazardous fluid returning from the surgical site because replacing or cleaning the Regulator after every surgical procedure would be prohibitively expensive and/or unduly burdensome. However, isolating the Regulator by placing fluid collection canisters between it and the surgical site is not desirable due to the costs of the canisters, the complexity of setting them up, the need to change them during the procedure when they become full, and the need to dispose of them at the end of the procedure.

(173) Referring to FIG. 52, to overcome the problems associated with the use of a Regulator, the system **100** may utilize a single-use or multi-use pressure regulator **5205** that is cost-effective to manufacture and disposable. The pressure regulator **5205** does not need to be isolated from the biohazardous fluids returning from the surgical site because of its disposability. The pressure regulator **5205** can be used in combination with a pressure source (e.g., an air pump) and one or more pressure sensors of the fluid management system **100** to sense and regulate the vacuum pressure provided by an external suction source to control the rate of fluid outflow from a surgical site and, thereby, assist in efforts to provide good distention and visualization. The pressure pump and pressure sensors may be included in an aspiration module **5201** that is configured to be operatively connected to the control system of the fluid management system **100**, or the pressure pump and pressure sensors may be integral to the main unit **102** of the fluid management system **100**. In embodiments that include the use of the aspiration module **5201**, inserting the pressure regulator **5205** into the aspiration module **5201** causes fluidic connections between the pressure regulator **5205** and the pressure sensing and gas bleed mechanisms and integrated pressure pump of

the aspiration module **5201**. After inserting the pressure regulator **5205** into the aspiration module **5201**, the user can manually connect an external suction source and the fluid return lines from the surgical site to connection ports **5202** (e.g., openings **5315**, **5317** shown in FIG. **53** and openings **5415**, **5417** shown in FIGS. **54-55**) of the pressure regulator **5205**.

(174) Referring to FIG. **53**, a first exemplary embodiment of the pressure regulator **5205** includes three chambers **5307**, **5309**, **5311** and a flexible membrane **5313**. The first chamber **5307** includes an opening or port **5315** for fluidly connecting to the external suction source. The second chamber **5309** has an opening or port **5317** for fluidly connecting to the surgical site (via one or more fluid lines or tubes). The third chamber **5311** includes one or more ports for connecting to a pressure source (e.g., pressure source **7449** of aspiration module **5201** shown in FIG. **74**) and a pressure sensor (e.g., pressure sensors **7451** of aspiration module **5201** shown in FIGS. **74-75**). In the illustrated embodiment, the third chamber **5311** has a first opening **5319** for connecting to the pressure sensor and a second opening **5321** for connecting to the pressure source.

(175) The flexible membrane **5313** is positioned to fluidly isolate (i.e., seal) the third chamber **5311** from both of the first chamber **5307** and the second chamber **5309**, which allows the pressure source and pressure sensor of the fluid management system **100** (which are connected to the openings **5319**, **5321** of the third chamber **5311**) to be fluidly isolated from biohazardous fluid returning from the surgical site and moving through the first and second chambers **5307**, **5309**. In some embodiments, a hydrophobic filter (not shown) is disposed between the flexible membrane **5313** and the openings **5319**, **5321** to provide further protection in preventing fluid from contacting the pressure source and pressure sensor of the fluid management system **100**. For example, the hydrophobic filter can prevent fluid from contacting the pressure source and regulator if the flexible membrane **5313** tears or ruptures.

(176) The first chamber **5307** is adjacent to the second chamber **5309** and separated from the second chamber **5309** by a substantially vertical extended member or wall **5323** and a substantially horizontal extended member or wall **5361**. The wall **5361** includes openings **5363** that fluidly connect the first chamber **5307** to the second chamber **5309**. The pressure source of the fluid management system **100** is configured to move the flexible membrane **5313** between an engaged position with the wall **5361** and one or more disengaged positions with the wall **5361**, where the first and second chambers **5307**, **5309** are fluidly isolated from each other when the flexible membrane is in the engaged position, and where the first and second chambers **5307**, **5309** are fluidly connected to each other (via openings **5363**) when the flexible membrane **5313** is in one of the disengaged positions. The size and spacing between of the openings **5363** prevent the membrane **5313** from rupturing due to over-extrusion through the openings when positive pressure is applied to stop flow across the valve. The size and spacing of the openings **5363** may vary based upon the elasticity and thickness of the material for flexible membrane **5313**. The number of openings **5363** can ensure adequate flow, and, in some embodiments, the total combined surface area of the openings on either side of each valve may be roughly equivalent to, or greater than, the inner cross-sectional area of the tubing expected to be attached to the port **5315**.

(177) The fluid management system **100** is configured to provide pressure to the third chamber **5311** through opening **5319**, and the system **100** is configured to sense and regulate the pressure within the third chamber **5311** by sensing pressure via the pressure sensor and opening **5321** and modulating the pressure provided by the pressure source to achieve the desired pressure setpoint (e.g., by modulating the air pump speed of pressure source **7449** of aspiration module **5201** shown in FIG. **74**). The system **100** controls the pressure source to cause the flexible membrane **5313** to stretch away from the wall **5361** when a more positive pressure exists in the chamber **5309** than the greater of the pressure in the first chamber **5307** or the pressure in third chamber **5311** (in addition to the force required to displace the flexible membrane **5313**). When a more positive pressure exists in the second chamber **5309** than in the first and third chambers **5307**, **5311** (in addition to the force required to displace the flexible membrane **5313**), fluid is able to displace the flexible membrane

5313 and create a fluidic connection between chambers **5307**, **5309** via the holes **5363** through the wall **5361** and the displaced flexible membrane **5313** such that the desired regulated pressure (e.g., the lesser of the pressure in the first chamber **5307** or the pressure in the third chamber **5311**, in addition to the force required to displace the flexible membrane **5313**) is supplied to the surgical site through the opening **5317** of the second chamber **5309**. In nominal conditions, the pressure of the external suction source present in the first chamber **5307** is the lower than the pressure of the other chambers **5309**, **5311** of the pressure regulator **5205**, the regulated pressure in the third chamber **5311** is greater than the pressure in the second chamber **5309**, and the regulated pressure in the third chamber **5311** is adjustable to allow regulation of the pressure supplied to the surgical site through the opening **5317** of the second chamber **5309**.

(178) When the first chamber **5307** and the second chamber **5309** are fluidly connected, the biohazardous fluid moves from the surgical site, into the second chamber **5309** through opening **5317**, through the opening between the flexible membrane **5313** and the wall **5361** via holes **5363** and into the first chamber **5307**, and through the opening **5315** to a waste collection of the system **100** or the facility. When the valve is desired to be closed and flow stopped from the surgical site, the system **100** applies a pressure more positive in the third chamber **5311** than the maximum pressure expected in the second chamber **5309** (e.g., the pressure caused by the weight of the water column in the height difference between the valve inlet **5317** and the surgical site) and the pressure in the chamber **5307**, in addition to the pressure required to displace the flexible membrane **5313**. When the pressure in chamber **5311** is greater than the pressure in both chambers **5307** and **5309** (in addition to the pressure required to displace the flexible membrane), then the flexible membrane **5311** is held with sufficient force against wall **5361** to counteract the other system pressures such that flow is substantially halted. The flexible membrane **5313** seals the third chamber from the first and second chambers **5307**, **5309** to prevent the biohazardous fluid from moving into the third chamber **5311** and contacting the pressure source and/or pressure sensors of the system **100**.

(179) Provided the pressure from the surgical site (as available in chamber **5309**) is a more positive pressure than the gauge pressure supplied by the external suction source through the first chamber **5307**, the gauge pressure supplied by the pressure source of the system **100** into the third chamber **5311**, and the pressure required to stretch flexible membrane **5313**, the regulated vacuum pressure supplied to the surgical site can be equal to the more-positive gauge pressure of the gauge pressure supplied by the external suction source (through the first chamber **5307**) or the gauge pressure supplied by the pressure source of the system **100** (into the third chamber **5311**) and the pressure required to stretch the flexible membrane **5313**. That is, provided the flow rate from the surgical site is negligible with respect to the flow capacity of the external pressure source (supplied through the first chamber **5307**), the pressure supplied to the surgical site (through chamber **5309**) will be the pressure closest to absolute vacuum of the pressure supplied by the external pressure source (through the first chamber **5307**) or the pressure supplied by the pressure source of the system **100** (into the third chamber **5311**) and the pressure required to stretch the flexible membrane **5313**.

(180) In certain embodiments, the pressure required to stretch the flexible membrane **5313** may be modeled by a transfer function to determine a pressure setpoint for the pressure source of the system **100** (supplied into the third chamber **5311**) that is required to achieve a desired regulated vacuum pressure in the second chamber **5309** that is supplied to the surgical site. The system **100** may be configured to vary the regulated vacuum pressure supplied to the surgical site, via the pressure sensor and pressure pump of the system **100**, by varying the pressure supplied to the third chamber **5311**. Also, the pressure supplied by the external suction source (through the first chamber **5307**) and supplied to the second chamber **5309** may be regulated to a more positive pressure by regulating the pressure provided to the third chamber **5311** by the pressure source of the system **100** to a greater pressure than supplied by the external suction source. Because the regulated pressure setpoint is variable, this also enables the pressure regulator **5205** to serve as a simple 2-way valve to enable and disable flow on demand by regulating the pressure supplied to the third

chamber **5311** with a pressure greater than the pressure in either the first chamber **5307** or second chamber **5309**.

(181) In certain embodiments, the flexible membrane **5313** is configured such that the pressure supplied by the pressure source in the third chamber **5311** causes the flexible membrane **5313** to stretch away from the wall **5361** and cause the regulated vacuum pressure supplied to the surgical site to be between about 10 mmHg and about 30 mmHg greater than the pressure provided by the pressure source of the system **100**. The flexible membrane **5313** can be made of, for example, neoprene, silicone, natural rubber, nitrile, EPDM, other rubber compounds, or any other material that allows the flexible membrane to be moved between the engaged and disengaged positions.

(182) The pressure regulator **5205** may have a housing **5325** that at least partially defines the three chambers **5307**, **5309**, **5311** and includes the openings **5315**, **5317**, **5319**, **5321**. The housing **5325** can be made of, for example, polycarbonate, any suitable type of plastic material, or any other suitable material. In certain embodiments, the housing **5325** has a first component **5327** that includes the first and second chambers **5307**, **5309**, and a second component **5329** that includes the third chamber **5311**, where the flexible membrane **5313** is positioned between the first and second components **5327**, **5329** to fluidly isolate the chambers of the first component **5327** from the chambers of the second component **5329**. The first component **5327**, the second component **5329**, and the flexible membrane **5313** can be connected by a snap-fit connection, an adhesive connection, one or more fasteners, laser welding, ultrasonic welding, vibration welding, or any other suitable means.

(183) Referring to the embodiment shown in FIG. **53**, if the lowest desired regulated pressure supplied to the surgical site is a positive gauge pressure, the pressure supplied by the external pressure source could be a positive or negative gauge pressure. If the desired regulated pressure is a negative gauge pressure (i.e., a vacuum pressure), then the pressure supplied by the external suction source may be required to be a negative gauge pressure that is more negative than the lowest gauge pressure desired to be regulated because pressure supplied by the external pressure source is not being sensed. In other words, the pressure supplied by the external pressure source (through the first chamber **5307**) is not required to be consistent (e.g., does not need to be regulated and may have pressure fluctuations) provided the highest gauge pressure supplied by the external pressure source is not a gauge pressure that is higher than the desired regulation pressure setpoint for the regulated source (e.g., the surgical site).

(184) Although the embodiment of the pressure regulator **5205** shown in FIG. **53** is effective for regulating an external suction source that provides a vacuum level that is known to be more-negative than the desired regulated vacuum pressure provided to the surgical site, a second embodiment of the pressure regulator **5205** (shown in FIGS. **54-73**) allows the fluid management system **100** to regulate an external suction source that is providing an unknown or variable vacuum level and sense when the vacuum level is sufficient to achieve the desired regulation setpoint. Referring to FIGS. **54** and **55**, the second embodiment of the pressure regulator **5205** utilizes two valves of the first embodiment (FIG. **53**) arranged in series.

(185) Referring to FIG. **54**, the second exemplary embodiment of the pressure regulator **5205** includes four chambers **5407**, **5409**, **5411**, **5412** and a flexible membrane **5413**. The first chamber **5407** includes an opening or port **5415** for fluidly connecting to the external suction source. The second chamber **5409** has an opening or port **5417** for fluidly connecting to the surgical site (via one or more fluid lines or tubes). The third chamber **5411** includes one or more ports **5419** for connecting to a pressure source (e.g., pressure source **7449** of aspiration module **5201** shown in FIG. **74**), and the fourth chamber **5412** includes one or more openings **5421** for connecting to one or more pressure sensors (e.g., pressure sensors **7451** of aspiration module **5201** shown in FIGS. **74-75**). Alternative embodiments can incorporate sensors from the same port connections of the pressure regulator **5205** for redundancy or improved regulation of the pressure source.

(186) The flexible membrane **5413** is positioned to fluidly isolate (i.e., seal) each of the third and

fourth chambers **5411**, **5412** from both of the first and second chambers **5407**, **5409**, which allows the pressure source and pressure sensor of the fluid management system **100** to be fluidly isolated from biohazardous fluid returning from the surgical site and moving through the first and second chambers **5407**, **5409**. The first chamber **5407** is adjacent to the second chamber **5409** and separated from the second chamber **5409** by a vertical extended member or wall **5423** and a horizontal extended member or wall **5465**. The wall **5465** includes openings **5467** that fluidly connect the first chamber **5307** to the second chamber **5309**. The flexible membrane **5413** is movable from an engaged position and one or more disengaged positions with the wall **5465**, where the first and second chambers **5407**, **5409** are fluidly isolated from each other when the flexible membrane **5413** is in the engaged position, and where the first and second chambers **5407**, **5409** are fluidly connected with each other (via openings **5467**) when the flexible membrane **5413** is in the disengaged position. The third chamber **5411** is adjacent to the fourth chamber **5412** and separated from the first chamber **5412** by a vertical extended member or wall **5431** and a horizontal extended member or wall **5461**. The wall **5461** includes openings **5463** that fluidly connect the third chamber **5411** to the fourth chamber **5413**. The flexible membrane **5413** is also movable from an engaged position and one or more disengaged positions with the wall **5461**, where the third and fourth chambers **5411**, **5412** are fluidly isolated from each other when the flexible membrane **5413** is in the engaged position, and where the third and fourth chambers **5411**, **5412** are fluidly connected with each other (via openings **5463**) when the flexible membrane **5413** is in the disengaged position.

(187) To better show that the embodiment of the pressure regulator **5205** shown in FIG. **54** utilizes two valves of the first embodiment of the pressure regulator shown in FIG. **53** in series, the first chamber **5407** is shown as having a first portion **5408** and a second portion **5410**, but the pressure across both the first and second portions **5408**, **5410**, as supplied by the external suction source, is identical (as there is no barrier capable of sealing the first and second portions **5408**, **5410** from each other). Referring to FIG. **55**, the first valve **5501** of the pressure regulator **5205** utilizes the external suction source (via the first portion **5408** of the first chamber **5407**) to regulate movement of the pressure supplied by the pressure source of the fluid management system from the third chamber **5411** to the fourth chamber **5412**. The second valve **5503** utilizes the pressure in the fourth chamber **5412** (via the movement of pressure from the third chamber **5411** to the fourth chamber **5412**) to regulate the movement of the pressure supplied by the external suction source from the second portion **5410** of the first chamber **5407** to the second chamber **5409** such that the pressure in the second chamber **5409** is substantially equal to the desired regulated vacuum pressure at the surgical site.

(188) Referring to FIGS. **54** and **55**, the pressure at the first portion **5408** of the first chamber **5407** causes the flexible membrane **5413** to be in either the engaged position or the disengaged position with the wall **5461**. For example, if the vacuum pressure supplied by the external suction source is more negative than the pressure supplied by the pressure source of the system **100** into the third chamber **5411**, the flexible membrane **5413** stretches away from the wall **5461** such that the third and fourth chambers **5411**, **5412** are fluidly connected. When the third and fourth chambers **5411**, **5412** are fluidly connected, the pressure in the fourth chamber **5412** is substantially equal to the pressure in the third chamber **5411**. Comparatively, if the vacuum pressure supplied by the external suction source is more positive than the pressure supplied by the pressure source into the third chamber **5411**, the flexible membrane is in the engaged position with the wall **5461** to fluidly isolate the fourth chamber **5412** from the third chamber **5411**.

(189) The fluid management system **100** senses the pressure within the fourth chamber **5412** via the one or more pressure sensors of the system **100**, which allows the system **100** to determine if the pressure supplied by the external suction source is not supplying sufficient vacuum pressure to meet the desired regulated vacuum pressure at the surgical site. That is, if the external suction source is not supplying enough pressure to cause the flexible membrane **5413** to stretch away from

the wall **5461**, air will slowly bleed out of the fourth chamber **5412** via a small orifice or valve of the system **100** that may be constantly or periodically opened to a more positive gauge pressure to gradually bring the pressure in the fourth chamber **5412** closer to this more positive gauge pressure, and the pressure sensor of the system **100** will sense that the pressure in the fourth chamber **5412** is not equal to the pressure supplied by the pressure source to the third chamber **5411**, which will cause the system to determine that the pressure supplied by the external suction source is not sufficient to meet the desired regulated vacuum pressure at the surgical site. The pressure sensed by the system **100** in the fourth chamber **5412**, given enough time to bleed off pressure, can be used to determine the actual pressure present in the first chamber **5407** if it is less than the positive gauge pressure that is slowly bled into the fourth chamber **5412**. If the system **100** determines that the external suction source is not supplying a sufficient vacuum pressure, the system **100** may be configured to notify the user to adjust the external pressure source (e.g., by increasing the suction setting of external pressure source, unclogging the line leading to the external pressure source, finding a leak in the line leading to the external pressure source, etc.) to ensure it is sufficient to down-regulate the vacuum pressure at the surgical site to the desired regulated vacuum pressure. (190) Referring to FIGS. **54** and **55**, the second valve **5503** of the pressure regulator **5205** utilizes the pressure supplied by the pressure source of the fluid management system in the fourth chamber **5412** to regulate the movement of pressure supplied by the external suction source to the second chamber **5409** and, consequently, the surgical site. That is, the system **100** controls the pressure source to cause the flexible membrane **5413** to stretch away from the wall **5465** to fluidically connect the first and second chambers **5407**, **5409** such that the desired regulated vacuum pressure is supplied to the surgical site through the opening **5417** of the second chamber **5409**. The biohazardous fluid then moves from the surgical site, into the second chamber **5409** through opening **5417**, through the openings **5467** between the flexible membrane **5413** the wall **5465** and into the first chamber **5407**, and through the opening **5415** to a waste collection of the system **100** or the facility. The flexible membrane **5413** seals each of the third and fourth chambers **5411**, **5412** from both of the first and second chambers **5407**, **5409** to prevent the biohazardous fluid from contacting the pressure source and/or pressure sensors of the system **100**.

(191) Provided the pressure from the surgical site as available in chamber **5409** is a more positive pressure than both the gauge pressure supplied by the external suction source (through the first chamber **5407**) and the gauge pressure supplied by the pressure source of the system **100** (into the third chamber **5411**) and the pressure required to stretch flexible membrane **5413**, the regulated vacuum pressure supplied to the surgical site can be equal to the more-positive gauge pressure of the gauge pressure supplied by the external suction source (through the first chamber **5407**) or the gauge pressure supplied by the pressure source of the system **100** (into the third and fourth chambers **5411**, **5412**) and the pressure required to stretch the flexible membrane **5413**. That is, provided the flow rate from the surgical site is negligible with respect to the flow capacity of the external pressure source (supplied through the first chamber **5407**), the pressure supplied to the surgical site (through chamber **5409**) will be the pressure closest to absolute vacuum of the pressure supplied by the external pressure source (through the first chamber **5407**) and the pressure required to stretch the flexible membrane **5413** or the pressure supplied by the pressure source of the system **100** (into the third and fourth chambers **5411**, **5412**) and the pressure required to stretch the flexible membrane **5413**.

(192) In certain embodiments, the pressure required to stretch the flexible membrane **5413** may be modeled by a transfer function to determine a pressure setpoint for the pressure source of the system **100** (supplied into the third and fourth chambers **5411**, **5412**) that is required to achieve a desired regulated vacuum pressure in the second chamber **5409** that is supplied to the surgical site. The system **100** may be configured to vary the regulated vacuum pressure supplied to the surgical site, via the pressure sensor and pressure pump of the system **100**, by varying the pressure supplied to the third and fourth chambers **5411**, **5412**. Also, the pressure supplied by the external suction

source (through the first chamber **5407**) and supplied to the second chamber **5409** may be regulated to a more positive pressure by regulating the pressure provided to the third and fourth chambers **5411**, **5412** by the pressure source of the system **100** to a greater pressure than supplied by the external suction source. Because the regulated vacuum pressure setpoint is variable, this also enables the pressure regulator **5205** to serve as a simple 2-way valve to enable and disable flow on demand by regulating the pressure supplied to the third and fourth chambers **5411**, **5412** with a pressure greater than the pressures in either the first chamber **5407** or second chamber **5409**.

(193) In certain embodiments, the flexible membrane **5413** is configured such that the pressure supplied by the pressure source in the third and fourth chambers **5411**, **5412** causes the flexible membrane **5413** to stretch away from the wall **5423** and cause the regulated vacuum pressure supplied to the surgical site to be between about 10 mmHg and about 30 mmHg greater than the pressure provided by the pressure source of the system **100**, such as about 20 mmHg greater than the pressure provided by the pressure source. The flexible membrane **5413** can be made of, for example, neoprene, silicone, natural rubber, nitrile, EPDM, other rubber compounds, or any other material that allows the flexible membrane to be moved between the engaged and disengaged positions.

(194) In certain embodiments, the fourth chamber **5412** is pneumatically attached to a small orifice or valve of the fluid management system **100** that may be constantly or periodically opened to bleed off pressure to a gauge pressure that is greater than or equal to the maximum gauge pressure expected from either the pressure source of the fluid management system **100** or the external suction source. The opened orifice or valve bleeds off pressure at a negligible flow rate compared to the flow rate capability of the pressure source of the fluid management system to ensure that the pressure inside of the fourth chamber **5412** is the greater gauge pressure of the external pressure source or the desired regulated pressure in the second chamber **5409** and any additional force required to stretch or open the flexible membrane **5413**.

(195) The embodiment of the pressure regulator **5205** shown in FIGS. **54** and **55** allows for non-wetted, indirect sensing of the pressure source of the fluid management system **100** to help ensure that the external suction source is of sufficient pressure to regulate to the desired vacuum pressure of the system's pressure source to down-regulate the vacuum pressure in the second chamber **5409** and at the surgical site. Being able to indirectly sense the pressure supplied by the external suction source via the cost-effective single use pressure regulator **5205** is beneficial because it enables the system **100** to prompt the user to adjust the external suction source (by increasing the suction setting of the external suction source, unclogging the line leading to the external suction source, finding a leak in the line leading to the external suction source, etc.) to ensure it is sufficient to down-regulate the vacuum pressure at the surgical site to the desired regulated vacuum pressure or otherwise prevent operation of the system if the regulated vacuum pressure levels would be detrimental to the safety or efficacy of the intended use of the regulated vacuum pressure from the system.

(196) FIGS. **56** through **64** show an embodiment of the pressure regulator **5205** shown in FIGS. **54** and **55**. In the illustrated embodiment, the pressure regulator **5205** includes a housing **5625** that at least partially defines the four chambers **5407**, **5409**, **5411**, **5412** and includes the openings **5415**, **5417**, **5419**, **5421**. The housing **5625** can be made of, for example, polycarbonate or any other suitable material. Referring to FIG. **58**, in certain embodiments, the housing **5625** has a first component **5827** that includes the first and second chambers **5407**, **5409**, and a second component **5829** that includes the third and fourth chambers **5411**, **5412**. The first and second components **5827**, **5829** can be, for example, injection molded pieces. The flexible membrane **5413** is positioned between the two components **5827**, **5829** to fluidly isolate the chambers **5407**, **5409** of the first component **5827** from the chambers **5411**, **5412** of the second component **5829**. In some embodiments, the pressure regulator **5205** may also include covers **5833** for covering the outer facing portions of the chambers for each component **5827**, **5829**. The first component **5827**, the

second component **5829**, the flexible membrane **5413**, and the covers **5833** can be connected by a snap-fit connection, an adhesive connection, one or more fasteners, ultrasonic welding, combinations thereof, or any other suitable means. In some embodiments, the pressure regulator **5205** may include one or more hydrophobic filters **5835** for helping maintain a bacterial barrier between the pressure source and pressure sensors of the fluid management system **100**.

(197) Referring to FIG. **64**, in certain embodiments, the pressure regulator **5205** shown in FIGS. **56-63** can be used in combination with an aspiration module (e.g., aspiration module **7404** shown in FIGS. **74-75**) that includes a pressure pump (e.g., pressure pump **7449** shown in FIG. **74**) and pressure sensors (e.g., pressure sensors **7451** shown in FIGS. **74-75**) to sense and regulate the external suction source to control the rate of fluid outflow from a surgical site. When the pressure regulator is inserted into the aspiration module, the pressure regulator **5205** may be configured to connect to a receiving mechanism **6437** (FIG. **64**) that automatically connects the port **5419** for the third chamber **5411** to a port **6439** that is operatively connected to the pressure pump of the aspiration module. The connection between the pressure regulator **5205** and the receiving mechanism may also automatically connect the port **5421** for the fourth chamber **5412** to a port **6441** that is operatively connected to sensors and or an air bleed mechanism of the aspiration module. The connection mechanism **6437** may include channels **6443** for receiving end portions **6445** of the pressure regulator **5205** to allow for a secure connection between the pressure regulator **5205** and the aspiration module. After inserting the pressure regulator into the aspiration module **5205**, the user can manually connect the external suction source to the port **5415** for the first chamber **5407** and manually connect the fluid return lines from the surgical site to the port **5417** for the second chamber **5409**.

(198) FIGS. **65** through **73** show another embodiment of the pressure regulator **5205** shown in FIGS. **54** and **55**. In the illustrated embodiment, the pressure regulator **5205** includes a housing **6525** that at least partially defines the four chambers **5407**, **5409**, **5411**, **5412** and includes the openings **5415**, **5417**, **5419**, **5421**. The housing **6525** can be made of, for example, polycarbonate or any other suitable material. In certain embodiments, the housing **6525** has a first component **6829** that includes the first and second chambers **5407**, **5409**, and a second component **6827** that includes the third and fourth chambers **5411**, **5412**. The first and second components **6827**, **6829** can be, for example, injection molded pieces. The flexible membrane **5413** is positioned between the two components **6827**, **6829** to fluidly isolate the chambers **5407**, **5409** of the first component **6827** from the chambers **5411**, **5412** of the second component **6829**. In some embodiments, the pressure regulator **5205** may also include covers **6633** for covering the outer facing portions of the chambers for each component **6827**, **6829**. The first component **6827**, the second component **6829**, the flexible membrane **5413**, and the covers **6533**, **6633** can be connected by a snap-fit connection, an adhesive connection, one or more fasteners, ultrasonic welding, combinations thereof, or any other suitable means. In some embodiments, the pressure regulator **5205** may include one or more hydrophobic filters **6635** for helping maintain a bacterial barrier between the pressure source and pressure sensors of the fluid management system **100**. The pressure regulator **5205** can also include one or more sealing members **6643** (e.g., O-rings) for making fluid tight connections. In certain embodiments, the housing **6525** has a gripping member or handle **6547** that helps a user insert and remove the pressure regulator from the aspiration module or other component of the fluid management system **100**.

(199) Referring to FIGS. **72** and **73**, in certain embodiments, the pressure regulator **5205** shown in FIGS. **65-71** can be used in combination with an aspiration module (e.g., aspiration module **7404** shown in FIGS. **74-75**) that includes a pressure pump (e.g., pressure pump **7449** shown in FIG. **74**) and pressure sensors (e.g., pressure sensors **7451** shown in FIGS. **74-75**) to sense and regulate the external suction source to control the rate of fluid outflow from a surgical site. When the pressure regulator **5205** is inserted into the aspiration module, the pressure regulator **5205** may be configured to connect to a receiving mechanism **7237** that automatically connects the port **5419** for

the third chamber **5411** to a port **6439** that is operatively connected to the pressure pump of the aspiration module. The connection between the pressure regulator **5205** and the receiving mechanism may also automatically connect the port **5421** for the fourth chamber **5412** to a port **7241** that is operatively connected to sensors and or an air bleed mechanism of the aspiration module. The connection mechanism **6437** may include channels (not shown) for receiving end portions of the pressure regulator **5205** to allow for a secure connection between the pressure regulator **5205** and the aspiration module (e.g., similar to as shown in the embodiment shown in FIG. **64**). After inserting the pressure regulator into the aspiration module **5205**, the user can manually connect the external suction source to the port **5415** for the first chamber **5407** and manually connect the fluid return lines from the surgical site to the port **5417** for the second chamber **5409**.

(200) FIGS. **74** and **75** illustrate an exemplary embodiment of the aspiration module **5201** that can be used with the fluid management system **100** shown in FIG. **1** and the various embodiments of the pressure regulators **5205** shown in FIGS. **53-73**. The aspiration module **5201** may include the receiving mechanism **7437** (e.g., receiving mechanism **6437** shown in FIG. **64** or receiving mechanism **7237** shown in FIG. **72**) for receiving the pressure regulator **5205**, an integrated pressure pump **7449**, one or more valves **7452**, **7453** for connecting the pressure pump to the pressure regulator **5205**, and a printed circuit board (PCB) **7448** having one or more pressure sensors **7451**.

(201) The integrated pressure pump **7449** may be configured to supply a positive pressure or a negative pressure to the pressure regulator **5205** via opening **5419** (FIG. **54**) of the pressure regulator. For example, the pump **7449** may be an air pump that includes two ports (not shown), and the aspiration module **5201** may include valves **7452**, **7453** (e.g., three-way valves) connected to the opening **5419** of the pressure regulator **5205** and the ports of the pump **7449**. A first port of the pump **7449** may be configured to pull air into the pump **7449**, and a second port of the pump **7449** may be configured to push air out of the pump, which allows the pump **7449** to supply both positive and negative pressures for pressure regulation and stoppage of flow through the valves. For example, if the port which pulls air into the pump **7449** is left open to ambient and the port which pushes air out of the pump **7449** is connected to a substantially sealed vessel, then the pump **7449** will build up positive pressure inside the substantially sealed vessel. Conversely, if the port which pushes air out of the pump **7449** is left open to ambient and the port which pulls air into the pump **7449** is connected to a substantially sealed vessel, then the pump **7449** will build up vacuum pressure inside the substantially sealed vessel. In this embodiment, the valves **7452**, **7453** are ported such that the pump ports can each be opened to ambient air pressure and connected to supply pressure to the aspiration module. Software of the control system may modulate the states of the valves **7452**, **7453** to configure the pump to supply positive or vacuum pressure to the system based on the desired valve state and regulation setpoint. The pump **7449** can be fluidly connected to the valves **7452**, **7453** by tubing. In other embodiments, the pump **7449** may be configured to supply either a positive pressure or a negative pressure to the pressure regulator, or the aspiration module **5201** may include separate positive and negative pressure pumps for supplying pressure to the pressure regulator **5205**. In certain embodiments, an accumulator **7450** is fluidly positioned between pressure pump **7449** and the pressure regulator **5205** to aid in regulating the pressure provided to the pressure regulator **5205** by the pressure pump **7449**.

(202) The one or more pressure sensors **7451** may be used to monitor both the pressure supplied to third chamber **5411** (FIG. **54**) of the pressure regulator **5205** by the pressure pump **7449** and a pressure within the fourth chamber **5412** (FIG. **54**) of the pressure regulator **5205**. In certain embodiments, the pressure sensors **7451** can be operatively connected to the accumulator **7450** by pneumatic tubing to monitor the pressure supplied to the pressure regulator **5205**, and the pressure sensors **7451** can be operatively connected to the opening **5421** (FIG. **54**) of the pressure regulator **5205** by pneumatic tubing to monitor the pressure within the fourth chamber **5412**. The pressure

sensors **7451** allow the fluid management system **100** to sense the pressure within the fourth chamber **5412** (FIG. 54) of the pressure regulator **5205** to determine if the pressure supplied by the external suction source that is connected to the first chamber **5407** (FIG. 54) is not supplying sufficient vacuum pressure to meet the desired regulated vacuum pressure at the surgical site. (203) Referring to FIG. 75, in certain embodiments, the aspiration module **5201** includes a valve **7554** that allows for the fourth chamber **5412** (FIG. 54) of the pressure regulator **5205** to be constantly or periodically opened to bleed off pressure to a gauge pressure that is greater than or equal to the maximum gauge pressure expected from either the pressure pump **7449** (FIG. 74) or the external suction source. The opened orifice or valve bleeds off pressure at a negligible flow rate compared to the flow rate capability of the pressure pump **7449** to ensure that the pressure inside of the fourth chamber **5412** (FIG. 54) is the greater gauge pressure of the external pressure source or the desired regulated pressure supplied to the surgical site.

(204) Referring to the operation of the fluid management system **100** discussed in the present application, the control system may be configured to guide the user through the setup process using, for example, instructions, illustrations, animations, video, and/or system feedback via the user interface **110**. Referring to FIGS. 76 and 77, in certain embodiments, the system **100** prompts a user (via the user interface **110**) to select the surgical discipline and procedure that will be performed, which can cause the system **100** to set default operating parameters for the procedure, as well as safe, permissible adjustment ranges for those parameters (as stored in a memory of the system **100**). For example, the system **100** may prompt a user to select a discipline of “Gynecology,” “Urology,” or “Orthopedic”; and, if the user selects “Urology,” the system **100** may prompt the user to select one of the following procedures: “Cystoscopy,” “PCNL,” “TURBT,” “TURP,” or “Ureteroscopy.” Based on the selection by the user, the system **100** may then set default operating parameters (e.g., pressure control mode or flow control mode, setpoint fluid pressure or flow rate, fluid warming condition enabled or disabled, fluid deficit monitoring enabled or disabled, etc.) for the procedure.

(205) In certain embodiments, the system **100** may provide instructions to a user for installing the tubing sets to the various components of the system **100**. For example, the system **100** may instruct the user to insert the cartridge assembly **419** (FIG. 4) that includes the fluid conditioner **420** (FIG. 4) and fluid warming cartridge **422** (FIG. 4) into the main unit **102**, and then place or route the tubing that connects the fluid supply containers and the cartridge assembly **419** into or through the pump **212** (FIG. 2). The system **100** may then prompt the user to indicate whether fluid deficit monitoring will be performed during the procedure. In certain situations, the system **100** may require fluid deficit monitoring be performed based on an input from the user as to the type of procedure that is being performed. For example, if the user selects an operative hysteroscopy procedure, fluid deficit monitoring is required. For other gynecological and urological procedures, fluid deficit monitoring may be optional. If the user did not select an operative hysteroscopy procedure and did not elect to enable fluid deficit monitoring for the selected procedure, the system **100** may instruct the user to spike and hang the fluid bags. If the user selected an operative hysteroscopy, or selected another procedure and elected to enable fluid deficit monitoring feature for the selected procedure, the system **100** may prompt the user to indicate whether one or more fluid types will be used during the procedure and what the fluid types are, as illustrated FIGS. 78 through 80.

(206) In various embodiments, the system **100** may be configured to monitor and display fluid deficit by fluid type. For example, in operative hysteroscopy, surgeons can utilize multiple fluid types during a procedure based on the type of procedure being performed and the surgical instruments employed. These fluids can differ in osmolality, electrolyte content, and viscosity. The amount of these fluids absorbed by the surgical patient depend on the fluid pressure, the length of the procedure, and the degree of surgical disruption of the venous sinuses in the endometrium and, importantly, the myometrium if the intrauterine fluid pressure is greater than the surgical patient's

mean arterial pressure. Thus, the capability of monitoring and displaying fluid deficit by fluid type enhances the safety for the patient.

(207) Referring to FIGS. **78-80**, after the user indicates the number of fluid types that will be used during the procedure, the system **100** may prompt the user (via the user interface **110**) to select the specific fluids will be used during the procedure. The system **100** may then set the maximum allowable deficit limit for each specific fluid type (based on information stored in a memory of the control system or based on information inputted by a user). For example, the maximum allowable deficit limit for hypotonic, electrolyte-free fluids may be 1000 ml, and the maximum allowable deficit limit for isotonic, electrolyte containing fluids may be 2500 ml. The system **100** may also set a maximum total deficit limit for the procedure based on a sum of the fluid deficits for the selected fluid types. For example, the maximum total deficit limit for the procedure may be 2,500 ml.

(208) In various embodiments, the system **100** will provide a user with instructions (via the user interface **110**) for hanging the fluid supply containers. For example, with regards to a first fluid type selected by the user, the system **100** may instruct the user as to which hanging members **116** (FIGS. **1** and **2**) to use, instruct the user to hang the fluid supply container(s) and then monitor the hanging member(s) **116** (FIGS. **1** and **2**) to confirm that the fluid supply container(s) were disposed on the correct hanging members, or prompt the user to indicate which hanging member(s) **116** (FIGS. **1** and **2**) will be used to hold the fluid supply container(s) and then monitor the weight of the hanging member(s) **116** (e.g., via load cells connected to the hanging members) to determine when the fluid container(s) are disposed on the corresponding hanging members **116**. The system **100** can then repeat the above process for the fluid container(s) holding the second fluid type.

(209) During the procedure, the system **100** may be configured to monitor and display the fluid deficit level for the first fluid type (via the user interface **110**) by subtracting an amount of fluid returned from the surgical site (e.g., as determined using the deficit cartridge **2010** and deficit module **104** or by monitoring a weight of fluid collection containers hanging from member(s) **116** (FIGS. **1** and **2**), etc.) from the volume of fluid pumped to the surgical site (e.g., as determined by monitoring the weight of the fluid supply containers, monitoring the amount of pump rotations of the pump **212**, etc.).

(210) In certain embodiments, the user can switch to the second fluid type via the user interface **110** (e.g., by pressing a “Change Fluid Type” button or other similar button). The system **100** may then instruct the user to close the tubing line(s) connecting the fluid supply containers for the first fluid type to the pump **212** (e.g., by closing clamp(s) on the tubing line(s)). The system **100** may also instruct the user to collect all residual fluid of the first fluid type from the surgical site, underbody drape, and the floor. Subsequently, the system **100** may instruct the user to fluidly connect the tubing lines of the fluid supply containers for the second fluid type (e.g., by opening clamp(s) on the tubing line(s)). In alternative embodiments, the system **100** may be configured to detect (via one or more processors of the control system) when a user switches to the second fluid type. In certain embodiments, the tubing lines may be automatically fluidly connected or disconnected by pinch valves.

(211) After fluidly connecting the tubing lines for the second fluid type, the user may initiate a purge of the first fluid type from the system **100**. In certain embodiments, the user will stop the pump, remove the scope and/or surgical instrument from the body cavity constituting the surgical site, and allow the cavity to drain the first fluid type into the underbody drape. When complete, the user will fluidly disconnect the first fluid type from the pump by closing the associated clamp(s), fluidly connect the second fluid type to the pump by opening the associated clamp(s), direct the scope and/or surgical instrument into the underbody drape and press a “Purge” or similar button on the user interface **110** of system **100** which will cause the system **100** to pump the volume of the second fluid type necessary to force the first fluid type from the cartridge assembly **419**, fluid inflow tube, and scope and/or instrument. In certain embodiments, system **100** may pump the

second fluid type necessary to cause the purge until the user presses a “Stop Purge” or similar button on the user interface **110** of system **100**. After the purge has been completed and the underbody drape has emptied, system **100** will record the fluid deficit for the first fluid type, empty the deficit cartridge **2010** (FIGS. **33-34**), and then indicate to the user via user interface **110** of system **100** that the procedure can proceed with the second fluid type. This process may be repeated to change back and forth between the first and second fluid types.

(212) After the first fluid type is purged from the system **100**, the system **100** may commence monitoring and display of the fluid deficit level for the second type of fluid. The system **100** may be configured to display the total fluid deficit, the deficit of the first fluid type, and/or the deficit of the second fluid type. In certain embodiments, a user may be able to elect, via a toggle switch or similar button on the user interface **100**, whether the system **100** displays the total fluid deficit, the deficit of the first fluid type, the deficit of the second fluid type, and/or any combination thereof.

(213) In some embodiments, if the user does not notify the system **100** of a fluid change (i.e., the change from the first fluid to the second fluid), the system **100** may stop the pump **212** to pause fluid flow. For example, the system **100** may prompt the user to indicate whether a change in fluid type was intended, and, if the user indicates that a change in fluid type was not intended, the system can instruct the user to check for any issues that may be affecting the weight on a hanging member associated with the other fluid type (e.g., such as leakage from the bag or an open or partially open clamp). If the user indicates that a change in fluid type was intended, the system **100** can instruct the user to initiate a purge of the system (as indicated above). The user may switch the fluids multiple times using the procedures described herein.

(214) In certain embodiments, a user that initially indicated only one fluid would be used in the procedure, may during the procedure, indicate to system **100** via user interface **110** that a second fluid will be used by pressing a “Settings” button or icon or similar button or icon on the user interface **110** and then pressing an “Add Second Fluid” button or icon or similar button or icon. The system **100** may then instruct the user via user interface **110** to hang the second fluid type, purge the first fluid type, record the deficit for the first fluid type, and instruct the user to continue the procedure using the second fluid type as indicated above. The system can then track the deficit for the first fluid type, the second fluid type, and total deficit.

(215) In certain embodiments, once the fluid containers have been placed on the hanging members **116**, the system **100** may guide the user to complete the tubing installation process. For example, if the fluid suction and collection module **106** is utilized, but the deficit module **104** is not utilized, the instructions can include connecting the fluid lines returning from the surgical site, underbody drape, and the floor (if applicable) to the fluid suction and collection module **106**. If the deficit module **104** is utilized, the instructions can include inserting the deficit cartridge **2010** into the deficit module **104**, connecting the suction source to the deficit cartridge **2010** (e.g., via vacuum port **2016**), and connecting the fluid return tubing lines (from the surgical site) to the deficit cartridge **2010** (e.g., via fluid inlet ports **2012**, **2014**). If the aspiration module **5201** is utilized, the instructions can include inserting the pressure regulator **5205** into the aspiration module **5201**, connecting the suction source to the pressure regulator **5205** (e.g., via port **5315** shown in FIG. **53** or port **5415** shown in FIGS. **54-55**), and connecting the fluid return tubing lines (from the surgical site) to the pressure regulator **5205** (e.g., via port **5317** shown in FIG. **53** or port **5417** shown in FIGS. **54-55**). If the fluid flow and evacuation module **5101** (FIG. **51**) is utilized, the instructions can include inserting the deficit cartridge **2010** (or similar single-use fluid volume monitoring cartridge) into the fluid flow monitoring and evacuation module **5101** and connecting the fluid return tubing lines (from the surgical site) to the deficit cartridge (e.g., via fluid inlet ports **2012**, **2014**).

(216) Following the tubing installation process, the system **100** may instruct the user to complete a priming process. For example, the system **100** may instruct the user to fluidly disconnect the fluid conditioner **420** (FIG. **10**) from the surgical instrument being used at the surgical site (e.g., by

closing a clamp on the tubing that connects the fluid conditioner **420** to the surgical instrument). The system **100** may also instruct the user to fluidly connect at least one of the fluid supply containers to the pump **212** (e.g., by opening a clamp on the tubing that connects the fluid supply container to the pump **212**). Subsequently, the system **100** may instruct the user to initiate priming of the tubing set (e.g., by pressing a “Prime” button or other similar button), which will cause the system **100** to pump fluid from the at least one fluid container and into the fluid conditioner **420** until the pressure sensors of the system **100** indicate that fluid in the fluid outlet chamber **1054** (FIG. **10**) of the fluid conditioner **420** has reached a certain fluid pressure, and/or until a fluid presence sensor (e.g., fluid presence sensor **948** shown in FIG. **9**) of the system **100** that targets the fluid outlet chamber **1054** indicates that the fluid has reached a certain level. After the system **100** determines that the pressure or fluid level in the outlet chamber **1054** is sufficient, the system **100** may stop the pump **212**. The fluid pressure within the outlet chamber **1054** may then be reduced by reversing the pump **212** or opening a solenoid valve (e.g., solenoid valve **951** shown in FIG. **9**) until the desired fluid pressure or fluid level in the outlet chamber **1054** of the fluid conditioner **420** has been achieved. The volume of fluid necessary to prime the tubing set may be known and added as a constant offset for the purposes of monitoring and displaying the volume of fluid pumped and the fluid deficit (if applicable).

(217) Referring to FIG. **81**, the system **100** may include a procedure run screen **8101** (via the user interface **110**). In the illustrated embodiment, the procedure run screen **8101** is shown for a system that is in pressure control mode. In alternative embodiments, the system **100** may be in flow control mode or flex control mode. Each of these control modes are discussed in more detail below. The user may start a procedure by pressing the “Play”, “Run”, or similar icon or button **8103** in the navigation bar at the bottom of the screen **8101**. The user may also, via adjustment control/buttons on the run screen **8101**, change default operating settings, such as, for example, the fluid pressure setpoint condition **8105** (if the system **100** is in a pressure control mode), the fluid flow rate setpoint condition (not shown-if the system **100** is in a flow control mode), and the deficit alarm level **8107** (if the deficit monitoring function is required or has been elected). In certain embodiments, the system **100** may only allow a user to change the default operating settings to be within the safe, permissible adjustment ranges **8111** for the procedure. The procedure run screen **8101** may also allow a user to enable/disable the fluid warming function by using switch **8113** and display the fluid temperature **8112**.

(218) In addition to displaying an actual condition **8115** and setpoint condition **8105** for pressure (or for flow if the system **100** is in a flow control mode), and an actual condition **8117** and setpoint condition **8107** for fluid deficit (if applicable), the procedure run screen **8101** may also display other information. For example, the procedure run screen **8101** may display fluid inflow or volume pumped **8121** to the surgical site, the fluid flow rate **8123** (if the system **100** is in pressure control mode), and/or the fluid pressure (not shown-if the system **100** is in flow control mode). The procedure run screen **8101** may also have a navigation bar consisting of icons or buttons that can be used before or during the procedure, such as, for examples, a “Settings” button **8128**, a “Help” or “Troubleshooting” button **8127**, a “Notifications” button **8129**, a “Maintenance” button **8181**, and/or an “End Case” or “End Procedure” button **8133**.

(219) In certain embodiments, pressing the “Settings” button **8128** brings up a settings screen (not shown) on the user interface **110** that allows the user to adjust, set, or enable other features of the system **100**. For example, referring to FIG. **82**, the system **100** may include a procedure settings screen **8202** that allows a user to set or adjust the control mode that the system **100** will follow for the procedure. In this embodiment, the system **100** may be set in a pressure control mode **8204**, a flow control mode **8206**, or a flex control mode **8208**. In certain embodiments, the system **100** may default to one of the control modes (e.g., the pressure control mode **8204**), but the user can change the type of control mode in the procedure settings screen **8202**. If the system **100** is in flow control mode **8206**, the system varies the speed of the pump **212** to achieve and maintain a user-selected

fluid flow rate setpoint (e.g., as set by the user on the procedure run screen **8101**) provided, however, that the maximum allowable fluid pressure for the procedure cannot be exceeded. In other words, the fluid pressure is varied to achieve the desired fluid flow rate.

(220) If the system **100** is in pressure control mode **8204**, the system **100** varies the speed of the pump **212** to achieve and maintain a user-selected fluid pressure setpoint (e.g., as set by the user on the procedure run screen **8101** shown in FIG. **81**) provided, however, that the maximum allowable fluid flow rate for the procedure cannot be exceeded. In other words, the fluid flow rate is varied to achieve the desired fluid pressure. In some embodiments, the system **100** can control to the user-selected fluid pressure setpoint as the desired fluid pressure at the system. In some embodiments, the system **100** can control to the user-selected fluid pressure setpoint as the desired fluid pressure at the surgical scope or instrument. Referring to FIG. **81A**, for ergonomic reasons corresponding to operating the graphical user interface **110**, hanging fluid supply containers from hanging members **116**, and inserting components (e.g., the cartridge assembly **419** and/or deficit cartridge **2010** described in the present application) into the main unit **102**, a height **H1** of main unit **102** may be higher than the surgical table **8151**. In certain embodiments, the system **100** can compensate for the head pressure resulting from the difference **H3** between the height **H1** of the main unit **102** and the assumed or inputted height **H2** of a patient **8153** in calculating the required system pressure. That is, the system **100** can equate the compensation height **H3** to a pressure adjustment, and then subtract the pressure adjustment from the user-selected fluid pressure setpoint at the surgical scope or instrument. In some embodiments, the system **100** can control to the user-selected fluid pressure setpoint as the desired fluid pressure in the body cavity constituting the surgical site. In these embodiments, in addition to calculating the pressure adjustment based on the compensation height **H3** described with reference to FIG. **81A**, the system **100** may also take into account known restrictions of the tubing set and assumed or calibrated restrictions of the surgical scope or instrument to determine the required system pressure. Accordingly, in some embodiments, the user may elect for the system **100** to monitor and display (via the graphical user interface **110**) the fluid pressure at the system, fluid pressure at the surgical scope or instrument, or fluid pressure in the body cavity constituting the surgical site.

(221) In endoscopic surgical procedures, good, steady distention and clear visibility are important to procedural efficacy and efficiency. Although fluid pressure and flow rate are the primary factors to achieve satisfactory surgical site distention and visibility, some users may lack a clear understanding of how fluid pressure and/or flow rate (as impacted by surgical site conditions and fluid inflow and outflow restrictions of the surgical instrument and the tubing sets delivering fluid to and from the surgical site) affect distention and visibility. Referring to FIG. **83**, in an exemplary embodiment, the system **100** may alternatively be set to a flex control mode that allows a user to achieve desired surgical site conditions simply by making distention and visualization adjustments. That is, in this mode, the user is not concerned with whether the system **100** is operating fluid pressure control mode or the fluid flow rate control mode, nor is the user concerned with the setpoint pressure or flow rate. Instead, the user may provide feedback to the system **100** regarding the surgical site conditions via the user interface **110**, and the system **100** determines whether to operate in pressure or flow control mode, as well as determines the proper setpoint for fluid pressure and/or flow rate for the procedure.

(222) Referring to FIG. **83**, if the user selects the flex control mode, the system **100** sets a default fluid pressure setpoint for the procedure and presents “Distention” controls or buttons **8351**, **8352** and “Visualization” controls or buttons **8353**, **8354** on the user interface **110**. For example, the user may increase distention by pressing the “+” (increase) button **8351** or may decrease distention by pressing the “-” (decrease) button **8352**, and the user may increase visualization by pressing the “+” (increase) button **8353** or may decrease visualization by pressing the “-” (decrease) button **8354**. The user interface **110** may also display other information (similar to the procedure run screen **8101** shown in FIG. **81**). For example, the user interface **110** may display fluid inflow or

volume pumped **8321** to the surgical site, the fluid flow rate **8323**, and/or the fluid pressure **8324**. The user interface may also display the fluid temperature **8312** and allow a user to enable/disable the fluid warming function by using switch **8313**. The user interface **110** may also have a navigation bar consisting of icons or buttons that can be used before or during the procedure, such as, for examples, a “Settings” button **8328**, a “Help” or “Troubleshooting” button **8327**, a “Notifications” button **8329**, a “Maintenance” button **8331**, and/or an “End Case” or “End Procedure” button **8333**.

(223) Referring to FIG. **84**, when the system is set to flex control mode (as shown at **8402**), the system **100** may default to pressure control mode (as shown at **8404**). In the illustrated embodiment, the adjustment of distention puts the system in fluid pressure control mode and adjusts the pressure setpoint for the procedure, and the adjustment of visualization puts the system in fluid flow rate control mode and adjusts the flow rate set point for the procedure.

(224) If a user adjusts the distention (as shown at **8406**), the system **100** maintains or transitions to the pressure control mode setting (as shown at **8408**) such that the system **100** can adjust the setpoint pressure of the procedure to meet the desired distention by the user. In various embodiments, distention adjustment to the fluid pressure setpoint can never exceed the maximum allowable setpoint pressure for the procedure. If the user indicates that additional distention is desirable, but the pressure setpoint is at the maximum allowable level (as shown at **8410**), the system **100** determines whether the flow rate for the system **100** is at the maximum allowable flow rate (as shown at **8412**). If the flow rate is at the maximum allowable flow rate, the system **100** can notify the user (via the user interface **110**) that the system **100** is operating at its pressure and flow rate limits for the procedure (as shown at **8414**). If the flow rate is not at the maximum allowable flow rate, the system **100** can instruct the user to open the scope inflow and outflow valves to increase the fluid flow rate (as shown at **8416**) and thereby increase the fluid moving through the surgical site. Going back to the step shown at **8410**, if the setpoint pressure of the fluid is not at the maximum allowable pressure, the system **100** determines whether the flow rate for the system **100** is at the maximum allowable flow rate (as shown at **8418**). If the flow rate is at the maximum allowable flow rate, the system **100** can instruct the user to restrict fluid outflow from the surgical site (as shown at **8420**) by partially closing the scope outflow valve, partially closing a clamp on the outflow tubing, and/or using a fixed or variable restrictor component in the outflow tubing of the surgical instrument. If the flow rate is not at the maximum allowable flow rate, the system **100** can increase the pressure setpoint (as shown at **8422**) to increase the distention.

(225) If a user adjusts the visualization (as shown at **8424**), the system **100** maintains or transitions to the flow control mode setting (as shown at **8426**) such that the system **100** can adjust the setpoint flow rate of the procedure to meet the desired visualization by the user. In various embodiments, visualization adjustment to the fluid flow rate setpoint can never exceed the maximum allowable setpoint flow rate for the procedure. If the flow rate setpoint is at the maximum allowable level (as shown at **8428**), the system **100** determines whether the fluid pressure for the system **100** is at the maximum allowable pressure (as shown at **8430**). If the pressure is at the maximum allowable pressure, the system **100** can notify the user (via the user interface **110**) that the system **100** is operating at its pressure and flow rate limits for the procedure (as shown at **8414**). If the pressure is not at the maximum allowable flow rate, the system **100** can instruct the user to restrict fluid outflow from the surgical site (as shown at **8432**), which allows the system **100** to increase the pressure of the fluid in the surgical site. The user can restrict outflow from the surgical site by, for example, partially closing an outflow valve of the surgical instrument, partially closing a clamp on the outflow tubing, and/or using a fixed or variable restrictor component in the outflow tubing of the surgical instrument. Going back to the step shown at **8428**, if the setpoint flow rate of the fluid is not at the maximum allowable flow rate, the system **100** determines whether the fluid pressure for the system **100** is at the maximum allowable flow rate (as shown at **8434**). If the pressure is at the maximum allowable pressure, the system **100** can instruct the user to open scope inflow and

outflow valves (as shown at **8436**) to increase the fluid flow rate through the surgical site. If the pressure is not at the maximum allowable pressure, the system **100** can increase the flow rate setpoint (as shown at **8438**) to increase the visualization.

(226) In other words, by utilizing the distention controls **8351**, **8352**, the user puts the system **100** in pressure control mode and adjusts the setpoint fluid pressure for the procedure up to the maximum allowable level for the procedure, while maintaining a maximum allowable flow rate which cannot be exceeded. By utilizing the visualization controls **8353**, **8354**, the user puts the system **100** in flow control mode and adjusts the setpoint fluid flow rate for the procedure up to the maximum allowable level for the procedure while maintaining a maximum allowable fluid pressure which cannot be exceeded. Accordingly, the system **100** provides the user more intuitive control over the surgical site conditions while at all times remaining within the safe pressure and flow rates for the procedure and displaying the actual fluid pressure and flow rate.

(227) Referring to FIGS. **85** through **87**, the system **100** may include a bolus feature or device that is used to temporarily increase the fluid pressure and/or flow rate to maintain or increase distention and/or to maintain or increase fluid flow for procedural and/or visualization purposes. The bolus feature or device may include a “Bolus” icon or button on one of the screens comprising the graphical user interface; pneumatic, electric, or wireless foot pedal; and/or other actuating device that allows a surgeon to temporarily increase the fluid pressure and/or flow rate for procedural and visualization purposes. The bolus device may be operatively connected to the pump **212** (FIG. **2**) of the system **100** such that a surgeon can activate the bolus device to temporarily increase the pressure or flow rate by operation of the pump **212**, and such that the pump returns to the normal setting for providing fluid at the setpoint pressure and flow rate after the surgeon deactivates the bolus device. Because the surgeon operates the bolus device, the bolus device can be provided when needed without requiring manual operation of a bolus device or the interaction of a circulating nurse with the system. The bolus feature also prevents the setpoint pressure and/or flow rate of the fluid management system to be changed for only a temporary increase in the pressure and/or flow rate.

(228) The bolus device may interface with the system pneumatically, electrically, or wirelessly (e.g., via Bluetooth). The bolus device can be configured by the user via the user interface **110**. For example, the user (via the user interface **110**) may use a toggle switch **8510** to change the bolus device between an “On” and “Off” state. In certain embodiments, the user may elect to have the bolus device operate in a momentary mode **8501** where the increase is sustained for as long as the foot pedal is pressed. The user may alternatively elect to have the bolus device operate in a maintained mode **8503** where the foot pedal is pressed to activate the bolus device and pressed again to deactivate the bolus device. In another embodiment, the user may elect to have the bolus device operated in a timed mode (not shown) where the foot pedal is pressed to activate the bolus device, and the system **100** maintains the bolus device in the activate state for a desired amount of time. Whether the increase caused by the bolus device is to the fluid pressure or flow rate setpoint may depend, for example, on whether the system **100** is in pressure mode or flow control mode at the time of bolus activation. The user may elect to have the increase equal a set increment (e.g., a 25 mmHg pressure increase or a 50 ml/min flow rate increase), a percentage increase over setpoint (e.g., 20%), or the maximum allowable fluid pressure or fluid flow rate for the procedure. The increase in fluid pressure and fluid flow rate may be limited to the maximum allowable setpoint for the procedure.

(229) The bolus device of the present application is beneficial because the fluid pressure or flow rate increase is known and safe. That is, the user sets the increase, the actual fluid pressure and flow rate are displayed on the user interface **110** of the system **100**, and the increase never exceeds the max allowable safe limit for the procedure. In addition, the duration of the increase is appropriate as determined and controlled by the surgeon. Also, because the surgeon controls the bolus device via, for example, a foot pedal, the bolus device does not need to be operated by a

circulating nurse, and the bolus device does not involve interacting with the system.

(230) In an alternative embodiment, the system **100** may include a temporary adjustment feature or device that is used to temporarily increase or decrease the fluid pressure and/or flow rate, which allows a user to temporarily increase or decrease distention and/or visualization at the surgical site. This adjustment feature or device may include a “Temporary” icon or button representing a temporary adjustment on one of the screens comprising the graphical user interface in combination with a foot pedal or other type of actuating device. In certain embodiments, the actuating device includes a pneumatic, electric, or wireless foot pedal(s) such as a foot pedal with rocker action or a dual foot pedal arrangement that allows a surgeon to temporarily increase or decrease the fluid pressure and/or flow rate for procedural and visualization purposes. The adjustment device may be operatively connected to the pump **212** (FIG. 2) of the system **100** such that a surgeon can activate the device to temporarily increase or decrease the pressure or flow rate by operation of the pump **212**, and such that the pump returns to the normal setting for providing fluid at the setpoint pressure and flow rate after the surgeon deactivates the adjustment device. Because the surgeon operates the adjustment device, the temporary adjustment to pressure and/or flow rate can be provided when needed without requiring manual operation of a device or the interaction of a circulating nurse with the system **100**. The adjustment device may interface with the system pneumatically, electrically, or wirelessly (e.g., via Bluetooth) and can be configured by the user via the user interface **110** for mode (momentary or maintained) and adjustment type (fixed, percentage, or max).

(231) In certain embodiments, the system **100** may include a printer (e.g., printer **218** shown in FIG. 2) for printing out pertinent information regarding a surgical procedure during or after the procedure. Referring to FIG. 88, in situations in which the system **100** is tracking a fluid deficit for the procedure, the user may elect (via the user interface **110**) to have the fluid deficit automatically recorded and printed out at set time intervals during the procedure (e.g., every 10 minutes). As illustrated in FIG. 88, the user interface **110** may include a toggle button to turn on or off the automatic recording and printing of fluid deficit at set time intervals during the procedure. This capability eliminates the need for the user to periodically check and manually record or print out the fluid deficit information. In some embodiments, the user may (via the user interface **110**) request to have the fluid deficit recorded at set time intervals during the procedure and printed at the end of the procedure. In some embodiments, the user may (via the user interface **110**) request that the deficit information from previous time intervals be displayed. In some embodiments, the system **100** may be configured to print the fluid deficit at volume intervals of the deficit monitoring (e.g., every 50 ml of volume for each fluid deficit individually and/or the total fluid deficit). In some embodiments, rather than printing to a printer of the system **100** itself, the system **100** may be configured to automatically communicate with a printer of the facility in which the system **100** is being used (e.g., via wire, Bluetooth, or WiFi) and print pertinent information at during or after the procedure, or at set time intervals during the procedure.

(232) The system **100** can be configured to offer the user the ability to printout pertinent procedure information at the end of the procedure, including, but not limited to, the date, procedure type, start time, end time, fluid volume pumped, fluid deficit (if applicable), fluid deficit at set time intervals (if applicable), average fluid pressure, fluid warming enabled/disabled, and/or average fluid temperature. In some embodiments, the user may be able to elect to have different or additional information printed, including, but not limited to, facility information, physician information, patient information, fluid deficit by fluid type (if applicable), fluid deficit by time increment (if applicable), fluid pressure range, fluid flow range, notifications and alerts list, and alarms list. Additionally, the user may elect to transmit pertinent procedure information via the Bluetooth or Wi-Fi capabilities of the system **100** to a data collection and/or record retention system of the facility.

(233) To avoid procedure interruption caused by depleted fluid supply bags, the system **100** may

record the initial weight of the fluid supply bag hung on each hanging member **116** (FIG. **1**), the current weight of the fluid supply bag on each hanging member **116**, and the current fluid flow rate for the procedure. The system **100** may also be able to provide audible and/or visual indicators if the system **100** determines that a fluid bag may become depleted. The system **100** may also provide an audible and visible indication when the estimated time before a fluid bag will become depleted has fallen below a specified level.

(234) Alternatively, referring to FIGS. **89** through **91**, the user may elect to receive audible and/or visual indicators (via the user interface **110**) if the percentage of fluid remaining (based on the initial volume of the fluid bag) falls below a specified level. Referring to FIG. **89**, the system **100** can be set to a “Time” setting **8902** that notifies a user when the amount of time until a bag becomes depleted drops below a predetermined amount of time (as shown at **8903**) based on the current or average fluid flow rate. The system **100** may, alternatively, be set to a “Percentage” setting **8904** that notifies a user when a percentage of the fluid supply remaining drops below a predetermine percentage (as shown at **8904**). Alternatively, the system **100** may be set to a “Volume” setting **8906** that notifies a user when a volume of fluid remaining in the fluid supply container drops below a predetermined volume (as shown at **8907**). Provided the above-mentioned levels are set appropriately, the user (typically a circulating nurse) has time to replace a fluid bag without an interruption to the procedure. To avoid confusion and ensure elevated levels of attention for alarm conditions, the user may mute or reduce the sound level of certain indicators, alerts, and alarms provided by switch or button **8910** and selection window **8912**. In some embodiments, however, adjustment of the alerts/indicators is not possible for certain safety critical alarms.

(235) Referring to FIGS. **92** and **93**, an exemplary embodiment of a fluid management system **9200** for a physician's office environment where gynecological and urological procedures are performed is shown. The system **9200** includes a main unit **9202** that may include any or all of the features described above for the main unit **102** of the fluid management system **100** used in an operating room environment. For example, the main unit **9202** may include a control system that has one or more processors (not shown) for controlling various components of the system **9200** (e.g., a user interface, and various fluid pressure sensors, vacuum pressure sensors, fluid temperature sensors, fluid presence sensors, etc.). The processors may execute instructions (e.g., software code) stored in a memory (not shown) of the system **100** and/or execute instructions inputted into the system by a user. In some embodiments, the control system may have “Bluetooth” capability for connecting to remotely located components or modules of the system **9200** and “Wi-Fi” capability for connecting to the internet. The control system may include a touch-screen graphical user interface **9210** for receiving one or more inputs from a user and displaying information of the system **9202** (e.g., information regarding fluid deficit, fluid temperature, fluid pressure, distention, visualization, etc.).

(236) The main unit **9202** may also include a pump (e.g., pump **212** shown in FIG. **2**) for fluid pressurization, a vacuum pump for providing suction, a fluid conditioning assembly (e.g., fluid conditioning assembly **315** shown in FIG. **3**) for receiving a fluid conditioner (e.g., fluid conditioner **420** shown in FIG. **10**) and sensing one or more characteristics (e.g., fluid presence, fluid temperature, etc.) of fluid moving through the fluid conditioner, hanging members **9216** (e.g., hooks) for hanging fluid supply or collection containers **9217** (e.g., bags, canisters, vessels, etc.). In some embodiments, the main unit **9202** may include a heating assembly (e.g., heating assembly **314** shown in FIG. **3**) for receiving a warming cartridge (e.g., warming cartridge **422** shown in FIGS. **14-18**) such that system can be configured for fluid warming during the procedure, if applicable. The processor of the control system can be in circuit communication with the pump, sensors, fluid conditioning assembly, heating assembly (if applicable), and hanging members **9216**; and the processor can be configured to control these components. In certain embodiments, the hanging members **9216** are operatively connected to load cells such that the control system can monitor a weight of the fluid containers **9217**.

(237) Although the system **100** for the operating room environment may include a cartridge

assembly **419** (FIG. 4) that includes a fluid conditioner **420** and a fluid warming cartridge **422**, the system **9200** for the physician's office may or may not include a fluid warming function. In embodiments that do not include a fluid warming function, the fluid conditioner **420** described with reference to the system **100** may also be used with the system **9200**, but the fluid conditioner **420** may include a connector or tube **841** (FIG. 8), a pulse damping component, or a channel integral to the fluid conditioner **420** that connects the inlet chamber **1053** (FIG. 10) to the outlet chamber **1054** (FIG. 10), rather than the warming cartridge **422**. If the system **9200** does include a fluid warming component, a warming cartridge (e.g., warming cartridge **422** shown in FIGS. 14-18) can be an accessory that attaches to the fluid conditioner **420** such that the system **9200** can perform fluid warming.

(238) The system **9200** may be configured to perform deficit monitoring by a weight-based method, as compared to the flow-based deficit monitoring method described with the fluid management system **100**. Flow-based deficit monitoring (as used with the system **100**) is appropriate for the operating room environment due to the generally higher fluid volume usage associated with the longer, more complex surgical procedures performed there. However, the complexity and cost of the flow-based deficit monitoring feature may not be necessary in a physician's office environment, as the surgical procedures performed there are generally shorter and use less fluid. Accordingly, the system **9200** may be configured to include weight-based deficit monitoring. In certain embodiments, the hanging members **9216** are configured for the dual purpose of holding and monitoring the weight of the fluid containers **9217**. That is, the processor of the system **9200** can be operatively connected to the load cells of the hanging members **9216**, which allows the system **9200** to monitor the weight of the fluid containers **9217**. At least one fluid container **9217** is for supplying fluid to the surgical site, and at least one fluid container **9217** is for fluid returning from the surgical site. The system **9200** monitors the weight of the hanging members **9216** to determine the fluid inflow volume to the surgical site (by based on the weight of the fluid supply container) and the fluid outflow volume returned from the surgical site (based on the weight of the fluid return container) to calculate the fluid deficit, which is the difference between the fluid inflow volume and the fluid outflow volume. The system **9200** can be configured to monitor and display the fluid volume and fluid deficit. The system **9200** can also be configured to provide a notification or alarm if the deficit level exceeds the default limit or the adjusted limit set by the user.

(239) Similar to the system **100** described in the present application, the system **9200** may be configured to guide the user through the setup process using instructions, illustrations, animations, and/or system feedback via the user interface **9210**. For example, the system **9200** may first prompt the user to select the surgical discipline and procedure that will be performed, which can cause the system **9200** to set the default operating parameters for the procedure as well as the safe, permissible adjustment ranges for those parameters.

(240) If deficit monitoring is required or elected by the user, the system **9200** can prompt the user to indicate the type of fluid that will be utilized and will set the maximum deficit limit for that fluid. The system **9200** can then instruct the user to hang a fluid supply container and indicate when the container has been placed on the hanging member **9216** (with the system confirming placement by monitoring the weight of each hanging member). The system **9200** can also instruct the user to hang the fluid supply container **9217** on a specific hanging member **9216** (with the system **9200** confirming placement by monitoring the weight of the designated hanging member). The system **9200** may then instruct the user to connect the fluid return lines to a fluid return container **9217**, connect the fluid return container **9217** to a suction source (e.g., an integrated suction source of the main unit **9200** or an external suction source), and then to hang the fluid return container on another hanging member **9216**. When the system **9200** senses that the fluid return container **9217** has been hung, it may set and record the empty fluid container weight to zero such that the system **9200** can properly calculate the fluid deficit for the procedure. Alternatively, the system **9200** can

instruct the user place fluid supply container **9217** on the hanging member **9216**, and when the system **9200** senses that a fluid supply container is properly placed, the system **9200** can instruct the user to prepare and hang a fluid return container as discussed above. The system **9200** can properly assign hanging members for each of the fluid supply containers and fluid return containers by monitoring the weight changes of the respective hanging members during the process, or by comparing the respective weight on the hanging members after the process has been completed. Although the system **9200** can accommodate standard canisters that can hold up to 5 L of fluid, the packaging of the tubing sets intended for procedures performed in the physician office environment can also be used for the described fluid collection function.

(241) If deficit monitoring is not required or elected, the system **9200** can prompt the user to place the fluid supply containers **9217** on the hanging members **9216**, place or route the tubing connecting the fluid containers(s) with the fluid conditioner (e.g., fluid conditioner **420**), into or through the pump **212**, and insert the fluid conditioner into the main unit **9202**.

(242) Following the tubing installation process, the system **9200** can then instruct the user to complete the priming process as described above with reference to the system **100**. When priming is complete, the user interface can transition to a procedure run screen (e.g., procedure run screen **8101**) where the user may start and control the procedure.

(243) While various inventive aspects, concepts and features of the disclosures may be described and illustrated herein as embodied in combination in the exemplary embodiments, these various aspects, concepts, and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein all such combinations and sub-combinations are intended to be within the scope of the present application. Still further, while various alternative embodiments as to the various aspects, concepts, and features of the disclosures—such as alternative materials, structures, configurations, methods, devices, and components, alternatives as to form, fit, and function, and so on—may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed. Those skilled in the art may readily adopt one or more of the inventive aspects, concepts, or features into additional embodiments and uses within the scope of the present application even if such embodiments are not expressly disclosed herein.

(244) Additionally, even though some features, concepts, or aspects of the disclosures may be described herein as being a preferred arrangement or method, such description is not intended to suggest that such feature is required or necessary unless expressly so stated. Still further, exemplary or representative values and ranges may be included to assist in understanding the present application, however, such values and ranges are not to be construed in a limiting sense and are intended to be critical values or ranges only if so expressly stated.

(245) Moreover, while various aspects, features and concepts may be expressly identified herein as being inventive or forming part of a disclosure, such identification is not intended to be exclusive, but rather there may be inventive aspects, concepts, and features that are fully described herein without being expressly identified as such or as part of a specific disclosure, the disclosures instead being set forth in the appended claims. Descriptions of exemplary methods or processes are not limited to inclusion of all steps as being required in all cases, nor is the order that the steps are presented to be construed as required or necessary unless expressly so stated. The words used in the claims have their full ordinary meanings and are not limited in any way by the description of the embodiments in the specification.

Claims

1. A fluid management system, comprising: a pump configured to deliver fluid from at least one fluid supply container to a surgical site; a control system for operating the pump, the control system

comprising: a user interface for communication between the control system and a user, including inputting data to the control system, and one or more non-contact sensors; and a fluid conditioning portion for receiving a fluid conditioner, wherein the fluid conditioner comprises a rigid body and a film defining an inlet chamber and an outlet chamber, wherein the fluid conditioner connects to the pump such that the inlet chamber receives the fluid delivered from the pump and the fluid moves through the fluid conditioner and out through the outlet chamber prior to moving to the surgical site, wherein the one or more non-contact sensors comprise a fluid presence sensor disposed proximate to the outlet chamber and configured to detect a presence of the fluid in the outlet chamber through the film, and wherein the control system is configured to expel excess air from the outlet chamber until a level of the fluid in the outlet chamber is detected by the fluid presence sensor such that an amount of air is maintained in the outlet chamber above a location of the fluid presence sensor.

2. The fluid management system of claim 1, wherein the fluid conditioner is aligned with the one or more non-contact sensors of the fluid management system when the fluid conditioner is disposed in the fluid conditioning portion.

3. The fluid management system of claim 1, wherein the fluid conditioner includes a port that allows a pressure sensor of the one or more non-contact sensors of the fluid management system to detect a pressure of the fluid.

4. The fluid management system of claim 1, wherein the fluid management system comprises the fluid conditioner and the fluid conditioner further comprises a connector for connecting the inlet chamber to the outlet chamber.

5. The fluid management system of claim 1, wherein the fluid management system comprises the fluid conditioner and the fluid conditioner includes an internal fluid path connecting the inlet chamber to the outlet chamber, wherein the internal fluid path is at least partially defined by the rigid body and the film.

6. The fluid management system of claim 1, wherein the fluid management system comprises the fluid conditioner and the fluid conditioner further comprises a fluid warming cartridge for fluidly connecting the inlet chamber to the outlet chamber, wherein the fluid warming cartridge is configured to expose the fluid to IR energy from a heating source of the fluid management system to heat the fluid when the fluid conditioner is inserted into the fluid management system.

7. The fluid management system of claim 6, wherein the fluid warming cartridge comprises a rigid body and flexible side sheets that define a conduit, wherein the conduit is fluidly connected to the inlet chamber and the outlet chamber such that the fluid received in the inlet chamber from the at least one fluid supply container moves through the conduit and into the outlet chamber, and wherein the conduit of the fluid warming cartridge is aligned with the heating source of the fluid management system when the fluid conditioner is disposed in the fluid conditioning portion of the fluid management system.

8. The fluid management system of claim 7, wherein the conduit comprises an inlet conduit and an outlet conduit, wherein the inlet conduit is defined by a first side of the rigid body of the fluid warming cartridge and a first flexible side sheet of the flexible side sheets, and wherein the outlet conduit is defined by a second side of the rigid body of the fluid warming cartridge and a second side sheet of the flexible side sheets.

9. The fluid management system of claim 7, wherein the flexible side sheets are configured to expand and contract as a pressure of the fluid moving through the conduit fluctuates to reduce pulsations of the fluid.

10. The fluid management system of claim 7, wherein the flexible side sheets are transmissive to the IR energy.

11. The fluid management system of claim 6, wherein the one or more non-contact sensors of the control system comprises an inlet temperature sensor that is aligned with the inlet chamber and an outlet temperature sensor that is aligned with the outlet chamber, and wherein the control system

- controls an amount of the IR energy provided to the fluid warming cartridge by the heating source based on at least one detected temperature by the inlet temperature sensor or the outlet temperature sensor and at least one of a default setpoint temperature of the fluid management system and a user inputted setpoint temperature.
12. The fluid management system of claim 11, wherein the control system is configured to disable the pump to prevent the pump from pumping the fluid when at least one of the inlet temperature sensor or the outlet temperature sensor detects a temperature that is greater than a predetermined value.
13. The fluid management system of claim 11, wherein the control system is configured to alert the user that the at least one of the default setpoint temperature of the fluid management system and the user inputted setpoint temperature is not achievable based on the at least one detected temperature by the inlet temperature sensor and the outlet temperature sensor.
14. The fluid management system of claim 11, wherein the one or more non-contact sensors of the control system comprise capacitive sensors.
15. The fluid management system of claim 11, wherein the one or more non-contact sensors of the control system further comprises an inlet fluid presence sensor that is aligned with the inlet chamber, and wherein the control system is configured to provide a first notification to the user through the user interface when the control system detects that the pump is operating and the inlet fluid presence sensor is not detecting the fluid in the inlet chamber.
16. The fluid management system of claim 15, wherein the control system is configured to provide a second notification to the user through the user interface when the control system detects that the pump is operating and the fluid presence sensor disposed proximate to the outlet chamber is not detecting the fluid in the outlet chamber.
17. The fluid management system of claim 1, wherein the one or more non-contact sensors of the control system comprises a pressure sensor connected to a port of the fluid conditioner such that the pressure sensor measures a fluid pressure in the outlet chamber, wherein the fluid presence sensor disposed proximate to the outlet chamber is a midpoint fluid presence sensor aligned with a midpoint of the outlet chamber, and wherein the control system is configured to expel the excess air from the outlet chamber when the pressure sensor is detecting a positive pressure in the outlet chamber and the midpoint fluid presence sensor is not detecting the fluid.
18. The fluid management system of claim 17, further comprising a solenoid valve connected to the port of the fluid conditioner that is movable from a closed position to an open position to expel the excess air from the outlet chamber.
19. The fluid management system of claim 17, wherein the one or more non-contact sensors further comprises a port fluid presence sensor aligned with the outlet chamber and located above the midpoint fluid presence sensor, and wherein the control system is configured to disable the pump to prevent the pump from pumping the fluid when the port fluid presence sensor detects the fluid.
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